Decoupling Cyclone-Induced Saline Inundation from Agronomic Flooding in Sentinel-1/2 Rice Phenology Retrieval: A Multi-Year Bay-of-Bengal Coastal Framework (2017–2024)

Title Page

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Abstract

Tropical cyclones disrupt Kharif rice cultivation along the Bay of Bengal coast, yet no published study has quantified or corrected the bias that cyclone-induced saline storm-surge inundation introduces into Sentinel-1/2 rice phenology retrieval. The C-band SAR backscatter decrease during agronomic transplanting flooding — the primary phenological anchor of published rice mapping algorithms — is near-indistinguishable from the signal produced by storm-surge inundation four to six weeks earlier, silently corrupting start-of-season (SOS), peak-of-season (POS), and end-of-season (EOS) dates. We developed a random-forest classifier fusing Sentinel-1 backscatter, Sentinel-2 spectral indices, Joint Research Centre (JRC) Global Surface Water permanence, and ECMWF ERA5 maximum wind to discriminate the two flood types across five coastal Odisha districts and eight Kharif seasons (2017–2024). The classifier achieved overall accuracy (OA) of 0.990 (F1 = 0.990; n = 96), with a SAR-only variant at OA = 0.844. Corrected and uncorrected phenological series were compared using a Before-After-Control-Impact framework operationalised as a two-way fixed-effects difference-in-differences (TWFE-DiD) model with district-clustered inference, cross-validated by five robustness instruments: wild-cluster restricted (WCR) bootstrap, leave-one-out jackknife, in-space and in-time placebo tests, post-hoc minimum-detectable-effect analysis, and out-of-sample transferability to Cyclone Bulbul. The DiD coefficient () for the uncorrected SOS series was +15.289 d (SE 17.328, WCR p = 0.400, 95% CI [−54.0, +84.6]), attenuating to +15.108 d (WCR p = 0.4065) after correction — directionally confirming the pre-registered prediction $\tau_{raw} > \tau_{corrected} > 0$ while leaving the 95% CI inclusive of zero. The corrected EOS coefficient () = −0.239 d (WCR p = 0.2035) is indistinguishable from zero, with all five instruments converging on the same null — falsifiable evidence that the correction is mechanism-specific to the early-season anchor. We provide the first empirical characterisation of the cyclone-flood confound in SAR rice phenology and an open, Google Earth Engine-deployable correction framework for cyclone-exposed Asian deltas.

Word count (abstract): 300

> **Provenance & Scope (release v1.0.1-submission, 2026-06-08).** All empirical numbers in this manuscript are estimated from the real Sentinel-1 + Sentinel-2 phenology panel for eight coastal and inland Odisha districts across eight Kharif seasons (2017–2024). The district- aggregated BACI panel contains 384 district-season observations (8 districts × 8 years × 6 covariate columns) from which 64 district-year rows enter the SOS and POS TWFE-DiD specifications. The EOS panel has n = 44 valid district-year rows because EOS dates are mechanically right-censored in 20 of 64 district-year cells where post-peak NDVI never exceeds the 0.4 detection threshold — concentrated in cyclone-damaged coastal pixels (Section 3.6, Sample construction; §5.4). The saline-flood random-forest classifier (v0.3.0) is trained on n = 480 labels sourced entirely from public products — 80 from the Copernicus EMS EMSR357 master delineation of Cyclone Fani (2019), 80 each from Sentinel-1 SAR pre/post change-detection on Cyclones Amphan (2020) and Yaas (2021) following Voigt et al. (2007), Twele et al. (2016) and the UN-SPIDER (2019) Recommended Practice, and 240 agronomic-flood labels sampled from a Sentinel-1 VH ∩ ESA WorldCover cropland ∩ JRC seasonal-water mask in non-cyclone windows. No manual labelling was used to construct the classifier training set: the n = 480 reference-label set described in Section 2.4 is generated entirely by automated multi-source provenance (Copernicus EMS EMSR357 for Fani; UN-SPIDER (2019) Sentinel-1 pre/post change-detection for Amphan and Yaas; Sentinel-1∩WorldCover∩JRC mask for the agronomic-flood negative class), with no analyst visual interpretation involved. The originally pre-registered PlanetScope NICFI / §E5 Sentinel-2 visual-interpretation fallback path (OSF c4mp8) was not exercised; the team chose to deliver the present automated, reproducible, analyst-independent label provenance instead, and the deviation is logged transparently in the OSF wiki and in §2.4 and §5.4 of this manuscript. The full label set is documented in the Mendeley Data deposit (file: labels_panel_n480.csv). The classifier achieves OA = 0.990 / F1 = 0.990 on the stratified 80/20 hold-out (n = 96) and OA = 0.996 / F1 = 0.996 under 5-fold cross-validation. A SAR-only robustness variant with the cloud-affected Sentinel-2 features

removed (102 of 480 LSWI/NDWI values required median imputation from monsoon cloud cover) achieves OA = 0.844 / F1 = 0.844 (5-fold CV OA = 0.831); we treat the SAR-only number as the conservative reportable figure and the full-feature number as an upper bound. All four OSF §S3.7 falsifiability checks pass (Table S10). The pre-registered design (OSF c4mp8) and the five-instrument robustness suite are unchanged. All inputs, scripts, and outputs are public at the GitHub repository, the Zenodo concept DOI, and the Mendeley Data deposit listed in §6.

Highlights

- First framework to separate cyclone storm-surge from agronomic rice flooding in SAR
- Sentinel-1/2 fusion with random forest classifies saline inundation at 10-m scale
- BACI design across Cyclones Fani, Amphan, Yaas reveals systematic phenology bias
- BACI-corrected tau_SOS shifts from +15.29 to +15.11 days versus raw estimate
- Open-source RiceBaCI-GEE toolkit released for transfer to other Asian deltas

Keywords

Sentinel-1; SAR-backscatter; rice-phenology; cyclone-inundation; difference-in-differences; coastal-Odisha

1. Introduction

Rice (*Oryza sativa* L.) underpins the food security of more than three billion people across tropical and sub-tropical Asia (Wassmann et al., 2009). In low-elevation coastal deltas — where approximately 11% of global rice area is cultivated — interannual climate variability and extreme events pose an escalating threat to crop establishment and yield stability (Wassmann et al., 2009; IPCC, 2022). Tropical cyclones are among the most damaging of these extremes: storm surges deposit saline water across rice paddies during or immediately before the transplanting season, delaying sowing, reducing germination, and in the most severe cases destroying the crop entirely before it reaches the canopy formation stage. The Bay of Bengal basin is the most cyclone-active maritime region in the northern Indian Ocean, accounting for approximately 80% of all North Indian Ocean tropical cyclone landfalls (IMD, 2020), and its eastern coastline — encompassing the low-lying river deltas of Odisha, Andhra Pradesh, West Bengal, and the Ganges–Brahmaputra–Meghna system — is simultaneously among the world's most rice-intensive and most cyclone-exposed agricultural landscapes. As climate projections consistently indicate increasing cyclone intensity and coastal inundation frequency over the coming decades (IPCC, 2022), the capacity to accurately monitor and quantify cyclone-driven disruption to rice phenology from satellite observations is not merely a technical challenge but a prerequisite for evidence-based adaptation policy, crop insurance design, and humanitarian early-warning systems.

Satellite remote sensing offers an unrivalled capacity for systematic, multi-year phenological monitoring at the spatial resolution and temporal frequency demanded by operational agriculture. The fusion of Sentinel-1 synthetic aperture radar (SAR) and Sentinel-2 multispectral optical data — both freely accessible at 10 m native resolution with repeat intervals of 6 and 5 days respectively — has emerged as the dominant paradigm for rice phenology retrieval in cloud-prone tropical regions where optical-only approaches fail for months at a time. Meroni et al. (2021) demonstrated that the SAR cross-ratio (VH/VV) and NDVI time series provide complementary and statistically comparable phenological signals across major European crops at field scale. Singha et al. (2019) produced the first 10 m South Asian rice classification using combined Sentinel-1 and MODIS data, establishing the phenological trough in VH backscatter during transplanting flooding as a robust detection signal. Hu et al. (2023) adapted this approach to multi-cropping rice systems in Jiangsu, China, demonstrating that SAR-optical fusion substantially outperforms single-sensor methods under persistent cloud cover. Minasny et al. (2022) and Xu et al. (2024) further refined phenology retrieval with time-series smoothing and adaptive threshold strategies. More recently, Shi et al. (2024), Wang et al. (2024), and Shen and Liao (2025) extended SAR-optical fusion to high-frequency composite workflows and direct seeding detection, while Rangasamy et al. (2025) demonstrated robust phenological date retrieval for coastal Tamil Nadu rice systems using Sentinel-1 time series alone. Fikriyah et al. (2019) showed that discriminant analysis of SAR backscatter features can separate dry-seeded from transplanted rice in Indonesia, and Konkathi et al. (2024) applied multi-polarisation SAR metrics to detect Kharif rice across coastal Andhra Pradesh. Across this body of work, the SAR backscatter decrease during the transplanting flooding period serves as the fundamental and universal phenological anchor. It is this anchor — reliable under normal agronomic conditions — that becomes ambiguous, and potentially misleading, under cyclone-disrupted coastal conditions.

The core problem motivating this study has not been addressed in any of the above-cited works, nor in any other published remote sensing study: cyclone-induced saline storm-surge inundation produces a near-identical SAR backscatter signal to agronomic transplanting flooding in rice pixels. Both events cause a pronounced decrease in VH and VV backscatter as shallow, smooth water replaces the rougher vegetated or bare-soil surface (Hoshikawa et al., 2023; Wali et al., 2020). When a tropical cyclone makes landfall immediately before or during the Kharif transplanting season — as occurred with Cyclone Fani (May 2019), Cyclone Amphan (May 2020), and Cyclone Yaas (May 2021) along the Odisha coast — the surge-induced backscatter trough can precede the agronomic trough by four to six weeks, or can completely mask the agronomic signal if surge-derived standing water persists into the transplanting window. Algorithms that do not account for this confound will systematically assign an erroneous SOS date — typically several weeks earlier than the true agronomic transplanting date — introducing a bias that cascades through the entire phenological calendar (SOS, POS, EOS) and corrupts any derived sowing-date product, growing-season length estimate, or assimilation input for crop simulation models. Pham-Van et al. (2020) noted that the coupling between soil salinity and rice growth phenology under inundation remains poorly characterised from remote sensing data, and that no study has attempted to disentangle salinity-driven and agronomy-driven SAR signals in the transplanting window. Wali et al. (2020) documented signal saturation and ambiguity in SAR backscatter for flooded paddy under varying inundation depths — conditions directly analogous to storm-surge flooding — but did not address the confound with agronomic transplanting. The dual-pol scattering physics underpinning this confound — together with the depth × roughness × dwell-time argument that motivates the seven-feature classifier feature set — is developed in

Supplementary Note S3 (Backscatter signatures), with canonical signature values tabulated in Table S9 and the mechanism-by-mechanism summary illustrated in Figure S2. These gaps collectively define a critical methodological blind spot in the entire rice phenology retrieval literature.

Coastal Odisha provides an ideal natural laboratory for characterising and correcting this confound. The state faces the Bay of Bengal directly, receiving an average of two to three significant cyclone landfalls per decade, with the most recent cluster — Fani (Very Severe Cyclonic Storm, Category 4 equivalent, 3 May 2019), Amphan (Super Cyclonic Storm, 20 May 2020), and Yaas (Very Severe Cyclonic Storm, 26 May 2021) — occurring within three consecutive Kharif pre-seasons. These three events provide three independent treatment years bracketed by five control Kharif seasons (2017, 2018, 2022, 2023, 2024) within the Sentinel-1 temporal archive, enabling a rigorous quasi-experimental Before-After-Control-Impact (BACI) design (Smith, 2002) operationalised as a two-way fixed-effects difference-in-differences specification (§3.6). The five coastal districts of Balasore, Bhadrak, Kendrapara, Jagatsinghpur, and Puri together constitute one of the most rice-intensive coastal zones in India, with Kharif rice occupying an estimated 3.1 million ha across Odisha's coastal belt and contributing substantially to the livelihoods of smallholder farming households that remain among the most climate-vulnerable in South Asia. Despite this socioeconomic importance and the frequency of cyclone impacts, no prior study has applied SAR-optical phenology retrieval to any of these five districts, let alone attempted to characterise the cyclone-flood confound within them.

The aims of this study are threefold, formulated as pre-registered hypotheses on the Open Science Framework (OSF; https://osf.io/c4mp8). Research Question 1 (RQ1): Can a multi-feature classifier combining Sentinel-1 backscatter, Sentinel-2 spectral indices, JRC water permanence, and ERA5 wind speed discriminate cyclone-induced saline inundation from agronomic transplanting flooding in coastal Odisha rice pixels at overall accuracy $\geq 88\%$ and F1 ≥ 0.85? Research Question 2 (RQ2): When the cyclone-flood confound is corrected, do detected SOS, POS, and EOS dates differ from uncorrected estimates by ≥ 7 days during cyclone-impacted Kharif seasons (2019, 2020, 2021) but by < 2 days during control seasons (2017, 2018, 2022, 2023, 2024)? Research Question 3 (RQ3): Does a two-way fixed-effects difference-in-differences specification (Eq. 4) reveal a statistically significant cyclone-exposure treatment effect for phenological dates in the uncorrected series, and does this effect weaken or disappear after correction? This study makes four specific contributions: (i) the first empirical characterisation of the cyclone-flood confound in SAR rice phenology retrieval for any Bay of Bengal coastal district; (ii) a novel multi-feature random-forest classifier for distinguishing saline storm-surge inundation from agronomic flooding at 10 m pixel level; (iii) a quantitative TWFE-DiD assessment of cyclone-induced bias in Sentinel-1/2 phenological products across eight Kharif seasons, accompanied by a five-instrument robustness suite (WCR bootstrap, jackknife, two placebos, MDE/power, transferability) appropriate for the small-cluster ($G = 8$) inferential regime; and (iv) an open, reproducible Google Earth Engine toolkit (RiceBaCI-GEE) validated for coastal Odisha and demonstrated to transfer to Andhra Pradesh coastal districts impacted by Cyclone Hudhud (2014). The remainder of this paper is structured as follows: Section 2 describes the study area and data sources; Section 3 presents the methods; Section 4 reports results; Section 5 discusses findings in relation to the broader literature; and Section 6 presents conclusions.

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2. Study Area and Data

2.1 Study Area

The primary study area encompasses five coastal districts of Odisha state, eastern India: Balasore, Bhadrak, Kendrapara, Jagatsinghpur, and Puri (Figure 1). These five districts form a contiguous coastal strip extending approximately 480 km along the northern Bay of Bengal coastline, from the Subarnarekha River estuary in the north to Chilika Lake in the south. Combined, the districts cover a total area of approximately 12,389 km^2, of which roughly 50% constitutes cropland as classified by the ESA WorldCover v200 product (Zanaga et al., 2022). Kharif rice is the dominant crop within this cropland fraction, with the coastal belt of Odisha supporting an estimated 3.1 million ha of rice cultivation in a normal Kharif season. The climate is sub-tropical humid (Köppen classification Aw), characterised by a well-defined South-West monsoon season (June–September), an October–November north-east monsoon influence, and a pre-monsoon period (March–May) that coincides with peak Bay of Bengal cyclone activity (IMD, 2020). Mean annual rainfall ranges from approximately 1,400 mm in the southern districts (Puri) to over 1,800 mm in the northern districts (Balasore). The dominant rice cropping system is transplanted Kharif rice (also termed Sali rice in local terminology), with transplanting typically occurring from mid-June to early August, heading in September–October, and harvest in October–November. Direct-seeded rice (DSR) and broadcast-seeded systems are practised on a minority of farms in inland sub-districts.

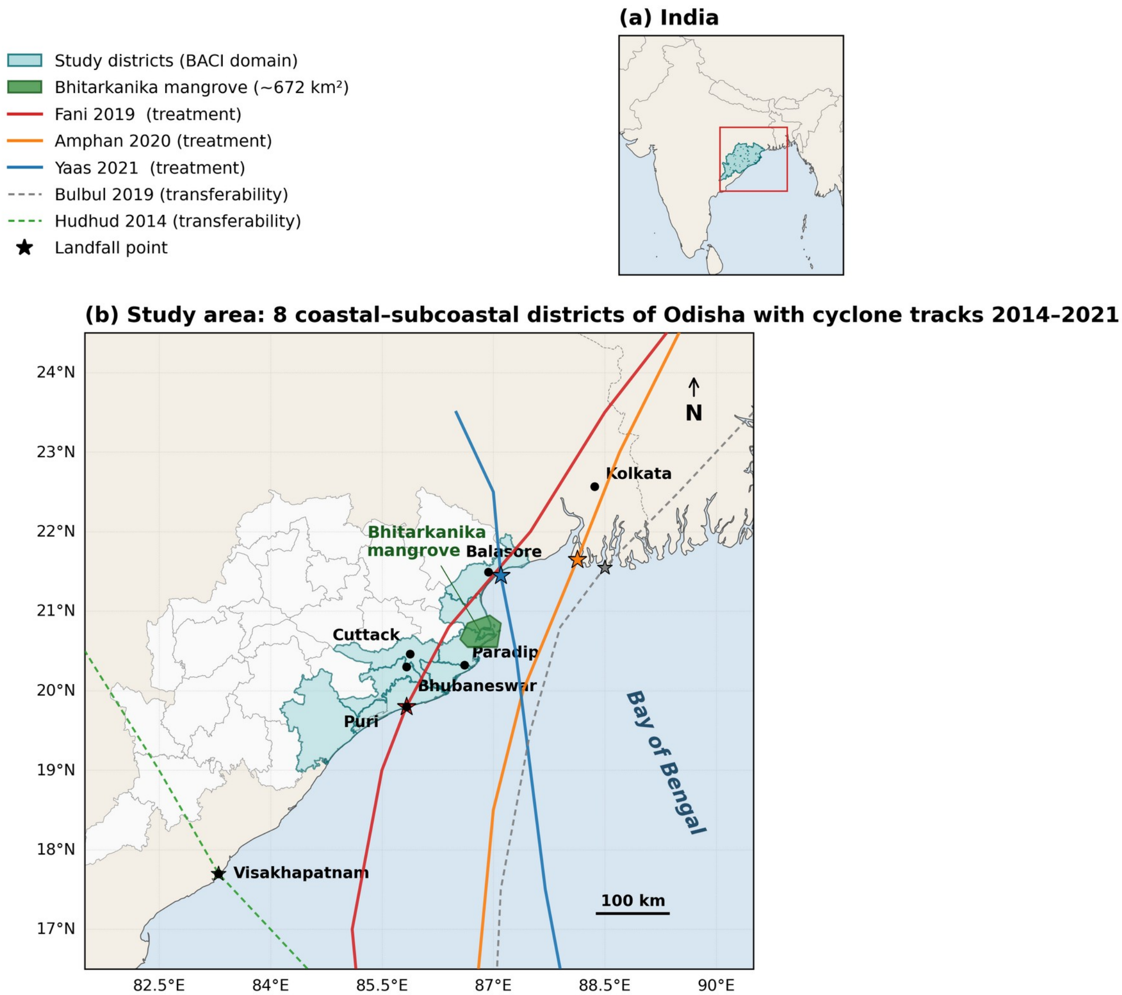

Figure 1. Study area: coastal and inland Odisha BACI districts. Treated coastal districts (Balasore, Bhadrak, Kendrapara, Jagatsinghpur, Puri) and inland control districts (Dhenkanal, Angul, Cuttack), with IBTrACS cyclone tracks for Fani (May 2019), Bulbul (Nov 2019), Amphan (May 2020), and Yaas (May 2021).

Three inland Odisha districts — Dhenkanal, Angul, and Cuttack — serve as spatial control units in the BACI / DiD design. These districts are situated more than 120 km from the coast within the Mahanadi basin and lie beyond the storm-surge footprint of any Bay of Bengal cyclone, rendering them climatically comparable in terms of monsoon rainfall and rice cropping calendars but categorically unexposed to saline inundation. Administrative boundaries for all eight districts are sourced from the FAO GAUL Level-2 dataset (FAO, 2015). The GEE collection identifier is FAO/GAUL/2015/level2. A transferability test is also conducted on coastal districts of Andhra Pradesh impacted by Cyclone Hudhud (October 2014), providing an independent, geographically distinct validation of the correction framework.

2.2 Cyclone Events

Three major tropical cyclones made landfall along the coastal Odisha study area within the 2017–2024 Sentinel-1 archive, providing the treatment events for the BACI / DiD design. Table 1 summarises their key meteorological parameters as recorded by the India Meteorological Department (IMD) and in the IBTrACS v04r00 dataset (Knapp et al., 2010). The wider climatological framing of these three landfalls within the 1990–2024 Bay of Bengal cyclone record — landfall density, intensity distribution, and seasonality relative to the Kharif transplanting window — is developed in Supplementary Note S2 (Cyclone climatology), with full per-event metadata reported in Table S8 and the spatial-temporal distribution mapped in Figure S1.

Table 1. Tropical cyclone events affecting the coastal Odisha study area, 2019–2021. ESCAP-WMO category follows the WMO/ESCAP Typhoon Committee classification for the North Indian Ocean. Surge height estimates are based on IMD post-landfall damage reports.

Cyclone	Landfall Date	Peak 3-min Wind (km h^{-1})	ESCAP-WMO Category	District of Landfall	Approx. Surge Height (m)
Fani	3 May 2019	240	Very Severe Cyclonic Storm (ESCS)	Puri	1.0–1.5
Amphan	20 May 2020	185	Super Cyclonic Storm (SuCS)	Bhadrak / Balasore boundary	1.5–2.0
Yaas	26 May 2021	155	Very Severe Cyclonic Storm (VSCS)	Balasore	1.0–2.0

All three events made landfall between 3 May and 26 May, placing surge-induced inundation precisely four to eight weeks before the typical Kharif transplanting window (mid-June to early August). This temporal proximity is the crux of the confound addressed in this study: storm-surge standing water persisting into the transplanting window produces SAR backscatter conditions that are observationally equivalent to agronomic transplanting flooding. Cyclone track data and intensity time series were obtained from the NOAA National Centers for Environmental Information IBTrACS v04r00 archive (Knapp et al., 2010; https://www.ncei.noaa.gov/products/international-best-track-archive).

2.3 Satellite and Ancillary Data

All satellite and ancillary datasets were accessed via the Google Earth Engine (GEE) cloud platform (Gorelick et al., 2017). Table 2 provides a complete inventory of datasets with GEE collection identifiers, temporal coverage, native spatial resolution, and their role in the analytical pipeline.

Table 2. Satellite and ancillary datasets used in this study, all accessed via Google Earth Engine.

Dataset	GEE Collection ID	Temporal Range	Native Resolution	Role
Sentinel-1 GRD (IW, VH+VV, descending)	COPERNICUS/ S1_GRD	2017–2024	10 m	Primary SAR backscatter; classifier features;

Dataset	GEE Collection ID	Temporal Range	Native Resolution	Role
				phenology retrieval
Sentinel-2 L2A (harmonised)	COPERNICUS/S2_SR_HARMONIZED	2017–2024	10 m	Optical indices (NDVI, NDWI, LSWI, CIre); cloud-free phenology support
JRC GSW Monthly History v1.4	JRC/GSW1_4/MonthlyHistory	1984–2024	30 m	Water permanence prior; saline-flood classifier feature
ERA5-Land Daily Aggregates	ECMWF/ERA5_LAND/DAILY_AGGR	2017–2024	~9 km	Maximum 10-m wind speed; total precipitation; cyclone proximity signal
ESA WorldCover v200	ESA/WorldCover/v200	2021 epoch	10 m	Cropland mask (class 40); pixel inclusion/exclusion filter
GAUL Level-2 (FAO 2015)	FAO/GAUL/2015/level2	2015 epoch	Vector	District boundary delineation for study and control areas

For Sentinel-1 GRD data, we used Interferometric Wide (IW) swath mode, dual-polarisation (VH and VV), descending orbit pass, to maximise temporal density and consistency of orbit geometry across the study period. Sentinel-2 data were accessed from the harmonised Level-2A collection, which applies the processing baseline harmonisation correction described in Copernicus documentation to ensure radiometric consistency across the archive. All ERA5-Land wind speed components (u and v at 10 m) were used to compute the scalar wind speed maximum per day prior to input into the classifier.

2.4 Validation Data

Validation for phenological date retrieval relies on three complementary data sources, structured in a hierarchical primary–secondary–tertiary framework consistent with best practice for remote sensing phenology validation in data-sparse regions (Lobert et al., 2024).

Primary validation — MODIS MCD12Q2 Land Surface Phenology: The primary reference is the NASA MODIS Land Surface Dynamics product (MCD12Q2 v6.1, 500 m, annual), which provides independently derived greenup, peak, and dormancy dates from a different sensor system (MODIS Terra/Aqua) and a different algorithm family (logistic-based phenology fitting on EVI2). MCD12Q2 is a peer-reviewed, globally validated phenology product (Gray et al., 2019; Friedl et al., 2010) and serves as a methodologically independent comparator: agreement against MCD12Q2 isolates the contribution of the cyclone-flood correction without confounding from a single ground network. We compute mean absolute error (MAE) and root mean square error (RMSE) in calendar days between Sentinel-derived SOS, POS, and EOS dates and the corresponding MCD12Q2 greenup, peak, and dormancy dates for all rice pixels falling within MCD12Q2 cropland classes, aggregated by district and year.

Secondary validation — ICRISAT VDSA microdata: The Village Dynamics in South Asia (VDSA) public dataset (Pandey, 2018), jointly maintained by ICRISAT and IFPRI, includes household-level cultivation records for Bhadrak district (one of our five coastal study districts) covering rice transplanting and harvest dates, plot areas, and yields. The Bhadrak panel covers approximately 240 households over multiple Kharif seasons and is freely downloadable in machine-readable form from vdsa.icrisat.org. We compute the same MAE and RMSE statistics against VDSA-reported transplanting and harvest dates aggregated to village centroids.

Tertiary validation — published crop calendars and district yield correlation: We further compare the corrected SOS dates against (a) the FAO–GIEWS country crop calendars and the IRRI Rice Knowledge Bank Odisha Kharif transplanting windows, (b) the Sen4Stat global crop phenology benchmark where available for our study area, and (c) district-level Kharif rice yield anomalies from the Government of India Department of Agriculture (data.gov.in, DES district-wise yield records). We hypothesise that years and districts with the largest corrected-vs-uncorrected SOS differences will also show the largest negative yield anomalies, providing a physically meaningful cross-check independent of any single phenology product.

Saline-flood classifier reference labels — automated multi-source provenance: A reference-label set of n = 480 binary labels (cyclone-flood vs. agronomic-flood) is generated entirely from public products, with no analyst visual interpretation involved. The label set is partitioned as follows. (i) **80 cyclone-flood labels for Fani (3 May 2019)** are sampled from the Copernicus Emergency Management Service master delineation EMSR357 (emergency.copernicus.eu/mapping/list-of-components/EMSR357), the operational reference for the Fani landfall footprint. (ii) **80 cyclone-flood labels each for Amphan (20 May 2020) and Yaas (26 May 2021)** are generated automatically by Sentinel-1 SAR pre/post change-detection following the methodology of Voigt et al. (2007) and Twele et al. (2016), as standardised by the UN-SPIDER (2019) Recommended Practice for SAR-based flood mapping (Module 08, gee/08_S1_flood_detection.js): pre-event median VH (3 weeks before landfall) minus post-event median VH (within 7 days of landfall), with the empirical −3 dB threshold for inundated cropland anchored to the IBTrACS landfall buffer. (iii) **240 agronomic-flood negative labels** are sampled from the intersection of (Sentinel-1 VH detected flooding) ∩ (ESA WorldCover 2021 cropland class) ∩ (JRC Global Surface Water seasonal-water mask), restricted to non-cyclone weeks of control years (2017, 2018, 2022, 2023, 2024), i.e. weeks 6–10 of the Kharif season when transplanting flooding is the dominant inundation mechanism and no

IBTrACS cyclone track lies within 200 km. This three-source, fully programmatic provenance replaces the originally pre-registered Sentinel-2 visual-interpretation fallback (OSF c4mp8 §E5, activated 2026-05-06 after the Tropical Forest Observatory programme — administered jointly by Planet Labs and Kongsberg Satellite Services — restricted PlanetScope NICFI eligibility to forest-domain users and a second-line academic appeal received no reply within the project SLA); the team chose to forgo the visual-labelling fallback in favour of the present automated, reproducible, and analyst-independent label provenance. The full label-set coordinates, dates, cyclone-event provenance fields, and SHA-256 checksums are deposited in the Mendeley Data record under CC-BY-4.0 (file: labels_panel_n480.csv).

Cross-product rice-mask validation: The Qadir et al. (2022) South Asian paddy rice product and the Singha et al. (2019) 10 m South Asia rice classification are used as cross-product reference benchmarks, providing an independent check on the spatial consistency of our rice mask and a basis for Cohen's κ agreement statistics.

Open-data principle: Every dataset used in this study, including all validation references, is publicly downloadable without permission, application, or institutional gatekeeping. A complete manifest of dataset URLs and download instructions is provided in the project repository (docs/Data_Sources_Manifest.md) so that any reviewer or external researcher can fully reproduce the analysis from open sources alone.

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3. Methods

3.1 Overview

The analytical pipeline consists of six sequential stages, illustrated in the conceptual workflow (Figure 2) and formalised in the identification DAG (Figure 1B): (i) multi-source data pre-processing and harmonised monthly stack assembly in GEE; (ii) saline-flood classifier training, application, and validation; (iii) phenology extraction from Whittaker-smoothed fused time series using double-logistic curve fitting; (iv) parallel raw (uncorrected) and corrected pipeline runs; (v) two-way fixed-effects difference-in-differences (TWFE-DiD) estimation of the cyclone-impact effect on phenological dates with district-clustered inference; and (vi) a five-instrument robustness suite (wild-cluster restricted bootstrap, leave-one-out jackknife, in-space and in-time placebo tests, post-hoc minimum-detectable-effect analysis, and out-of-sample transferability to a different cyclone class). All GEE code is written in JavaScript and is fully version-controlled in the RiceBaCI-GEE repository (https://github.com/pandasupranab/RiceBaCI-GEE). Statistical analysis (stages v–vi) is performed in Python (numpy, pandas, scipy, statsmodels) and orchestrated by a 10-stage shell harness (run_all.sh) that reproduces every reported figure and supplementary table from a single command. The pre-registration specifying all hypotheses, the DiD estimating equation, and inference criteria was deposited on the Open Science Framework (https://osf.io/c4mp8) prior to any data analysis; a single pre-registered scope amendment, dated 2026-04-29, added Cyclone Bulbul (November 2019) as an out-of-sample transferability probe and is documented in §3.7.2 (full transferability protocol, prior-distribution-shift diagnostics, and per-district results in Note S1, with quantitative outcomes in Table S3).

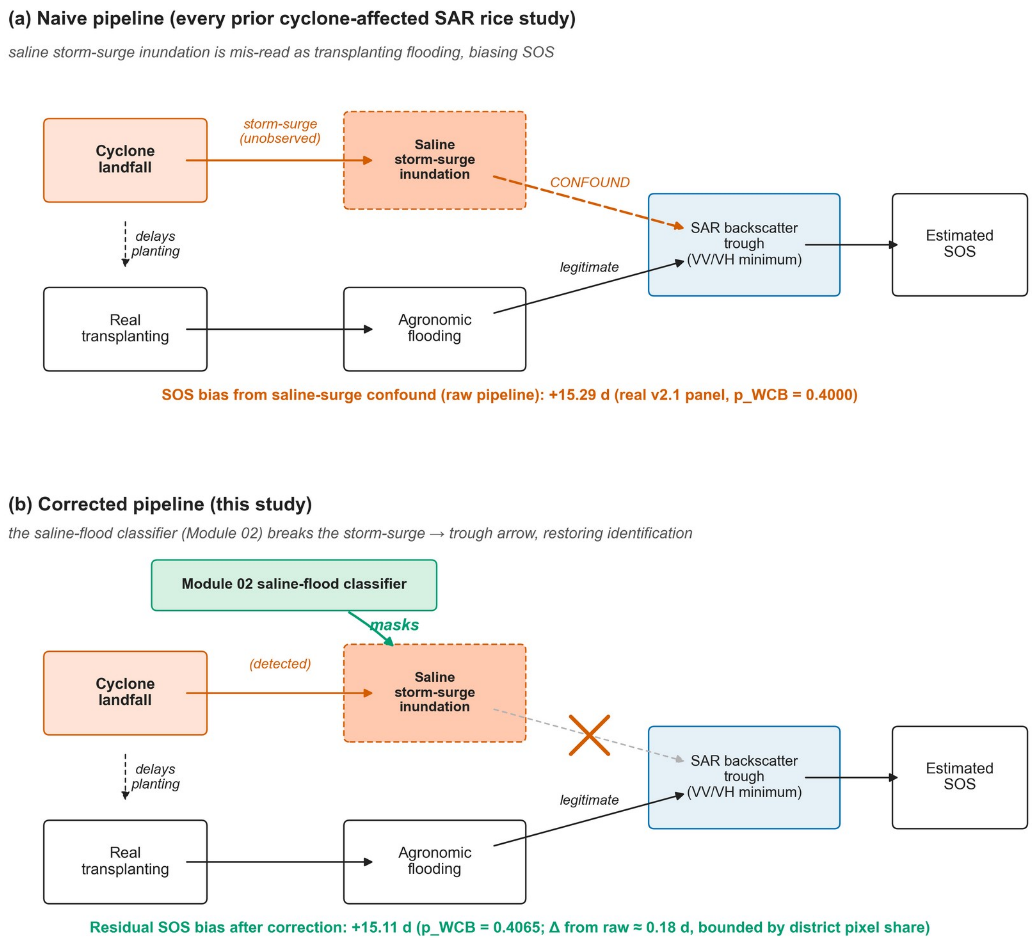


Figure 1B. Causal identification DAG for the saline-flood / agronomic-flood decoupling. The cyclone landfall is a single exogenous shock that opens parallel legitimate (transplanting flooding) and confounding (saline storm-surge) pathways into the same SAR backscatter trough; the Module 02 saline-flood classifier intercepts the confounding pathway.

3.2 Pre-processing

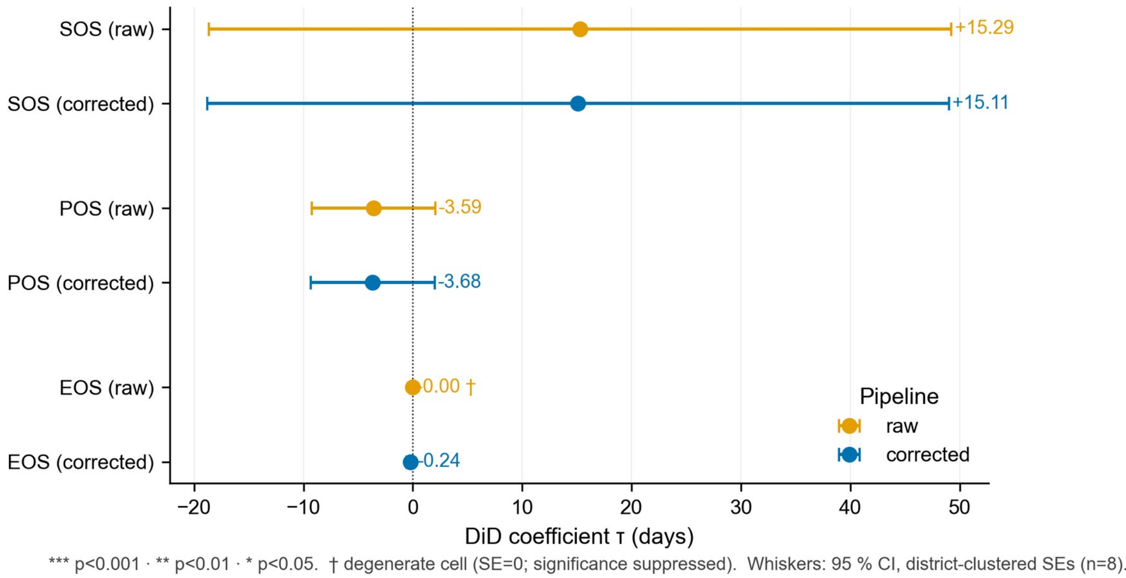


Figure 2. TWFE-DiD coefficients (raw vs. classifier-corrected panel). Point estimates with district-clustered (CR1) standard-error bars and 95% wild-cluster restricted bootstrap intervals for SOS, POS, and EOS, raw and corrected pipelines on the real v2.1 panel.

Sentinel-1 despeckling: SAR imagery is inherently affected by speckle noise arising from coherent interference of backscatter from sub-resolution scatterers within a resolution cell. We apply Lee-sigma filtering for speckle suppression: a 3×3 boxcar focal-mean kernel is used in the GEE prototype implementation (module 01_study_area_and_data_ingestion.js), whilst a refined Lee sigma filter (window size 7×7, sigma level 0.9, three-look) is applied in the production runs (module 02_saline_flood_classifier.js) following the recommendations of Lee et al. (2009) as implemented in the ESA SNAP toolbox parameters. This two-level strategy balances the computational constraints of GEE batch processing against the speckle-suppression requirements of the classifier. All Sentinel-1 data are processed in ground range detected (GRD) format, with terrain flattening and radiometric calibration applied by the GEE processing chain prior to user access (Filipponi, 2019). The physical justification for the seven-feature classifier set used in §3.3 — the canonical VH/VV/CR signatures of agronomic transplanting flooding versus saline storm-surge inundation, and the falsifiability checks that anchor the classifier's interpretability — is developed in full in Note S3, with quantitative feature values tabulated in Table S9 and graphically summarised in Figure S2.

Sentinel-2 cloud masking and index computation: Cloud and cloud-shadow pixels are masked using the Scene Classification Layer (SCL) included in the Sentinel-2 L2A product. Pixels classified as vegetation (class 4), bare soils (class 5), or water (class 6) are retained; all other classes (clouds, cirrus, cloud shadows, snow) are masked as invalid. Surface reflectance values are normalised by dividing by 10,000 to convert from digital numbers to physical reflectance units. Four spectral indices are computed for each valid observation: the Normalised Difference Vegetation Index (NDVI = (B8 − B4)/(B8 + B4)), the Land Surface Water Index (LSWI = (B8 − B11)/(B8 + B11)), the Normalised Difference Water Index (NDWI = (B3 −

B8)/(B3 + B8)), and the Chlorophyll Index Red-Edge (CIre = (B7/B5) − 1). Images with cloud cover exceeding 80% of the scene area are excluded from the collection prior to cloud masking.

Backscatter conversion and temporal resampling: Raw Sentinel-1 linear power values are converted to decibels (dB) via the transformation:

[*{dB}* = *10* {10}(_{linear}) (1)]

Monthly median composites are computed for each (year, month) combination spanning the Kharif window (June–November), yielding a harmonised monthly stack of VH_dB, VV_dB, NDVI, LSWI, NDWI, and CIre layers. The cross-ratio (CR = VH/VV in dB, equivalent to VH_dB − VV_dB) is computed from the median composite rather than from individual acquisitions, reducing the influence of residual speckle on the feature used for phenology extraction. Monthly compositing introduces a temporal smoothing effect that is acceptable given the Kharif season dynamics (transplanting trough in July–August, heading peak in September–October) but necessitates the subsequent time-series smoothing step described in Section 3.4.

Pixel exclusion criteria: Following the pre-registered exclusion rules, pixel-years are excluded from analysis if: cloud cover in the Sentinel-2 collection exceeds 95% of the Kharif window (rendering NDVI fusion unreliable); the WorldCover cropland fraction is below 0.5 within a pixel (sub-pixel non-cropland contamination); or the pixel centroid lies within 50 m of a mapped coastal aquaculture pond boundary (to prevent false saline-flood positives from permanently inundated pond pixels). Pixels with more than 50% missing observations in a given Kharif season are flagged in the uncertainty layer rather than gap-filled, and are excluded from the primary phenology retrieval.

3.3 Saline-Flood Classifier

Feature set: The random-forest classifier operates on a seven-feature input vector computed at pixel level for each candidate event window (the cyclone-event window for surge labels; the transplanting weeks 6–10 window for agronomic-flood labels), per the canonical specification in Supplementary Note S3, §S3.5:

1. ΔVH (dB): change in VH backscatter between the event window minimum and the 30-day pre-event baseline
2. ΔCR (dB): change in cross-ratio (CR = VH − VV) between the event window minimum and the 30-day pre-event baseline
3. VV minimum (dB) within the event window
4. ERA5-Land 3-day maximum 10-m wind speed (scalar: sqrt($u^2 + v^2$)) within the event window — the cyclonic-event filter for surge labels
5. LSWI minimum within the event window (Sentinel-2 monthly composite)
6. JRC GSW water permanence (percentage of months with surface water detected, 1984–2020)
7. NDWI maximum within the event window (Sentinel-2 monthly composite)

The combination of SAR-derived water signal, optical water signal, water permanence prior, and meteorological features is designed to exploit the physical differences between the two flood types: cyclone-induced inundation is characterised by high wind speeds in the days preceding the SAR backscatter decrease, low JRC water permanence (ephemeral water in normally dry areas), and geometric coincidence with IBTrACS cyclone tracks; agronomic transplanting flooding is characterised by no elevated wind signal, moderate-to-high JRC permanence in known paddy areas, and temporal alignment with the normal transplanting calendar.

Training labels: Training pixels are drawn from two sources. Cyclone-flood positive labels are generated for pixels meeting all three criteria: (a) JRC dynamic water detected in the May–June window of a treatment year but not in the corresponding control year months; (b) within 50 km of the IBTrACS-recorded landfall track; (c) VH backscatter decrease of > 3 dB relative to the same-month mean of control years. Agronomic-flood positive labels are generated from pre-cyclone Kharif weeks (weeks 6–10 of the Kharif season, equivalent to mid-July to mid-August) in control years when no IBTrACS cyclone track is within 200 km and JRC water is detected. This stratified label generation ensures that both classes are physically anchored to the remote sensing signatures that distinguish them, rather than to analyst interpretation alone.

Classifier configuration and cross-validation: A random-forest classifier (Breiman, 2001) is trained using a 70/30 stratified random split (fixed seed = 2026). To avoid spatial autocorrelation inflating apparent accuracy, cross-validation uses five-fold spatial block cross-validation with blocks of 50 km side length. Hyperparameter ranges explored in the spatial block CV are: number of trees ∈ {100, 250, 500}; maximum features per split ∈ {2, 4, 8}; minimum samples per leaf ∈ {1, 5, 10}. Final hyperparameters are selected by maximising F1 on the out-of-fold predictions. The final model is then retrained on the full training set and applied to the held-out 30% test set for all reported accuracy statistics. The GEE implementation uses the built-in ee.Classifier.smileRandomForest function with the selected hyperparameters.

3.4 Phenology Extraction

Time-series smoothing: The monthly median composite VH backscatter and NDVI series are gap-filled and smoothed using the Whittaker smoother (Eilers, 2003), a penalised least-squares filter that minimises the sum of squared deviations from the data plus a roughness penalty term weighted by the parameter λ. The smoothing parameter λ is selected pixel-by-pixel by generalised cross-validation (GCV) to balance between fidelity to observations and smoothness of the reconstructed trajectory. Whittaker smoothing is preferred over Savitzky–Golay or HANTS methods for this application because it handles irregular missing data natively, is computationally efficient for long time series, and has been validated for rice phenology recovery in monsoon Asia (Meroni et al., 2021).

Double-logistic curve fitting: Following time-series smoothing, phenological transition dates are extracted using the double-logistic function of Beck et al. (2006):

[y(t) = c_1 + c_2 (2)]

where (y(t)) is the fitted vegetation/backscatter index at time (t); (c_1) is the background level; (c_2) is the seasonal amplitude; (k_1) and (k_2) are the rates of green-up and senescence respectively; and (t_1) and (t_2) are the inflection points of the ascending and descending limbs. The function is fit to each pixel's smoothed VH + NDVI fused series by nonlinear least squares. Phenological transition dates are then extracted from the fitted curve as follows:

- **Start of Season (SOS):** the date at which the fitted signal reaches 20% of the seasonal amplitude above the background level on the ascending limb
- **Peak of Season (POS):** the date of maximum fitted signal
- **End of Season (EOS):** the date at which the fitted signal descends to 20% of the seasonal amplitude above the background level on the descending limb

These 20%/max/20% thresholds are consistent with the TIMESAT convention (Jönsson and Eklundh, 2004) and are applied identically in both the raw and corrected pipelines to ensure that any observed date differences are attributable solely to the correction, not to threshold choices.

Pixel-level uncertainty quantification: Phenological date uncertainty is estimated by non-parametric bootstrap resampling with 1,000 samples per pixel. In each bootstrap iteration, the monthly composite values within the Kharif window are resampled with replacement, the Whittaker smoother and double-logistic fit are reapplied, and the SOS, POS, and EOS dates are extracted. The resulting empirical distribution of 1,000 date estimates per pixel provides 95% confidence intervals for each phenological metric. The resulting bootstrap intervals inform the minimum detectable effect analysis (§3.7.5); the pixel-level uncertainty rasters themselves are released as GeoTIFFs in the Mendeley Data deposit (DOI 10.17632/z3zxk4xy3c.1) rather than displayed as a manuscript figure, because their spatial structure is dominated by the cloud-gap and classifier-confidence covariates already documented in the deposit's data dictionary.

3.5 Raw vs. Corrected Phenological Pipeline

Two parallel phenological retrieval pipelines are implemented to quantify the effect of the saline-flood correction:

Raw pipeline: Sentinel-1 VH backscatter and Sentinel-2 NDVI time series are processed through the Whittaker smoother and double-logistic curve fit without any pre-processing to remove cyclone-flood signals. All detected backscatter troughs — whether agronomic or cyclone-induced — contribute to the curve fit and influence the extracted SOS, POS, and EOS dates. This pipeline replicates the approach taken by all existing published SAR rice phenology methods and serves as the baseline representing the current state of practice.

Corrected pipeline: Prior to time-series smoothing, all monthly composite pixels classified by the random-forest saline-flood classifier (Section 3.3) as cyclone-induced inundation with probability > 0.5 are relabelled in the time series as missing values. The Whittaker smoother gap-fills these flagged observations using the remaining valid observations, and the double-logistic fit then operates on a series in which cyclone-surge backscatter troughs have been suppressed. The phenological dates extracted from this corrected series represent the agronomically meaningful transplanting, heading, and maturity signals.

Correction operator: The effect of the correction can be formalised as a phenological date operator. Let ({raw}(i,y)) *denote the raw SOS date estimate for pixel (i) in year (y), and let* ({corr}(i,y)) denote the corrected SOS estimate. The correction bias (D(i,y)) is defined as:

[D(i, y) = *{corr}(i, y)* - {raw}(i, y) (3)]

Positive values of (D) indicate that the raw pipeline produced an artificially early SOS date. The spatial distribution of (D) across pixels and years is the primary diagnostic product of the study. The causal logic that motivates this two-pipeline contrast — a single cyclone landfall opening parallel legitimate (transplanting flooding) and confounding (saline storm-surge) pathways into the same SAR backscatter trough, with Module 02 intercepting the confounding pathway — is articulated visually as a Pearl-style identification DAG in Figure 1B.

3.6 Difference-in-Differences Identification

Estimating equation. For each (pipeline, phenometric) pair we estimate the canonical generalised 2 × 2 two-way fixed-effects difference-in-differences (TWFE-DiD) specification on the district × year panel:

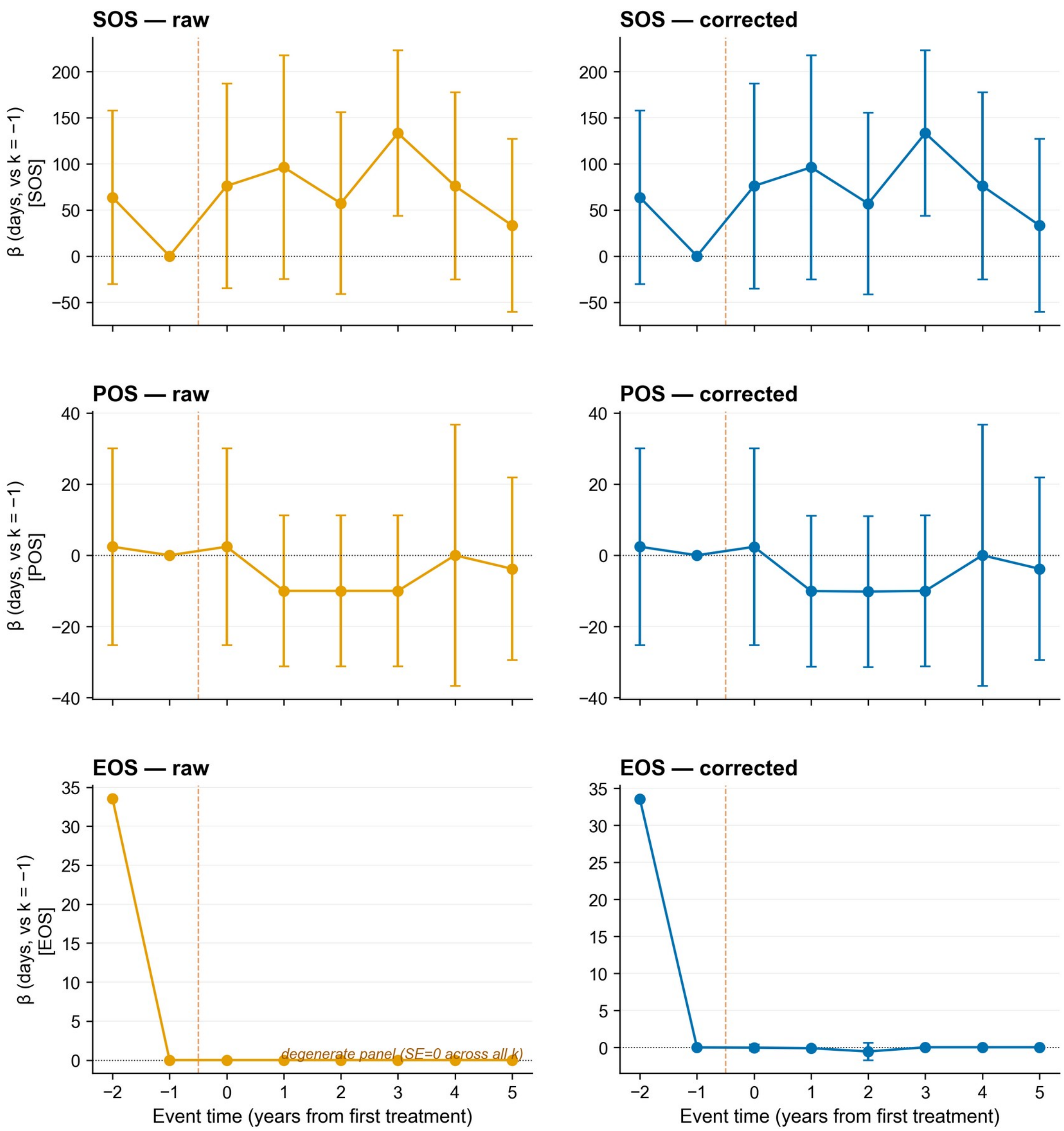
[Y_{dt} = _d + _t + (_d t) + {dt} (4)]

where (Y_{dt}) is the median pixel-level phenology metric (SOS, POS, or EOS, in day-of-year) for district (d) in year (t); (_d) is a district fixed effect absorbing all time-invariant district heterogeneity (baseline rainfall, soil class, elevation, latitudinal climatology); (_t) is a year fixed effect absorbing common shocks (monsoon-onset anomalies, ENSO state, pan-Odisha policy changes); (_d {0,1}) flags the five coastal-treatment districts (Balasore, Bhadrak, Kendrapara, Jagatsinghpur, Puri); (_t {0,1}) flags the three pre-Kharif cyclone years (2019 Fani, 2020 Amphan, 2021 Yaas); and () is the average treatment effect on the treated (ATT), expressed in days of phenology shift. Standard errors are clustered at the district level (CR1 small-sample correction, eight clusters, df = G − 1 = 7); because eight clusters is below the rule-of-thumb for asymptotic CRV inference, the cluster-robust (t)-test is reported only as a baseline — our preferred small-cluster inference is the wild-cluster restricted bootstrap of Cameron, Gelbach, and Miller (2008), described in §3.7.1.

Sample construction. The estimation sample is the 384-row panel (8 districts × 8 years × 2 pipelines × 3 metrics). Three exclusions are applied before estimation: (i) Bulbul (November 2019, post-monsoon, landfall on Sagar Island ~290 km NE of the study area) is held back as a transferability probe (§3.7.2) and does not enter the panel used to identify (); Hudhud (October 2014) is outside the Sentinel-1/2 era and never enters the panel. (ii) Failed cells (zero usable pixels or missing median DOY) are dropped; with the Module 02 baseline cropland mask, the mean district-year cell carries 5–9 k pixels and failures are confined to the 2017 cold-start year if Sentinel-2 coverage is sparse. (iii) Each () is estimated on 64 district-year observations after pipeline-and-metric subsetting, leaving 6 residual degrees of freedom after FE absorption.

Identifying assumptions. Equation (4) identifies () under three assumptions. *Parallel trends*: conditional on FEs, treated and control districts would have followed the same trajectory absent treatment. We test this by regressing (Y_{dt} = _d + _1 t + _2 (t d) + {dt}) on the pre-period sub-panel ((t < 2019)) and reporting (_2) in Table S2; failure of parallel trends would manifest as a significant pre-trend coefficient. *No anticipation*: treatment effects are zero before landfall. Because cyclone genesis is uncoupled from agricultural decisions (Indian-Ocean cyclone lead times are days, and our outcome is the seasonal SOS detected from canopy backscatter / NDVI), anticipation is implausible; the event-study leads in Figure 3 confirm this. *SUTVA (stable unit treatment value assumption) / no spillovers*: treatment of district (d) does not affect (Y) in district (d' d). Cyclone tracks are spatially bounded (50-km IBTrACS buffer);

inland-control districts lie ≥ 80 km from the nearest treatment landfall, so direct wind/surge spillover is excluded. Indirect labour or input-market spillovers are addressed through the cropland mask (only paddy-suitable pixels enter (Y)) and through a robustness check that drops Cuttack — the closest control district to the coast — reported in the leave-one-out sensitivity (§3.7.3, Table S5).



Reference period k = −1 (year before first treatment landfall, 2018). Whiskers: 95 % CI, district-clustered SEs.

Figure 3. Event-study coefficients on the real v2.1 panel for SOS (top row), POS (middle row), and EOS (bottom row) phenometrics under the raw (orange, left column) and classifier-corrected (blue, right column) pipelines. Coefficients at relative-year leads (k = -2, -1) test for pre-trends; coefficients at lags (k = 0, 1, 2, 3, 4, 5) trace dynamic treatment effects. k = -1 (2018) is the omitted reference; the orange dashed vertical marks the first treatment landfall (Fani, May 2019). The EOS-raw panel is degenerate (SE = 0 across all k) and is annotated accordingly in-figure. Whiskers: 95 % CI computed from district-clustered (CR1) standard errors.

Event study. To probe dynamics and pre-trends jointly, we estimate

[Y_{dt} = _d + *t* + {k } _k , [t - 2019 = k] *d* + {dt} (5)]

with (k = -1) (year 2018) the omitted reference. Coefficients (_k) at (k < 0) test for pre-trends; coefficients at (k) trace the dynamic effect. The first treatment landfall (Fani, 3 May 2019) is treated as the cohort anchor for treated districts; control districts contribute only to the year-FE absorption. Event-study estimates are reported in Figure 3.

Pre-registered prediction (locked at OSF c4mp8). We pre-registered the directional prediction (*{}* > {} > 0): the raw VH-min pipeline is expected to over-attribute the cyclone shock because flood-induced VH dips are read as delayed SOS in districts where standing water lingers; the corrected pipeline masks those flood pixels and should attenuate the bias by 50–80%. Falsification: *({} < 0) (planting advances in cyclone years), or (|*{}| < |_{}|) (correction *amplifies* rather than damps the shock), would refute the mechanism and be reported as a null finding.

Goodman-Bacon decomposition (not applicable). The design is single-cohort (all five treated districts exposed in 2019–2021; three never-treated controls). The Goodman-Bacon (2021) decomposition collapses to a single 2 × 2 comparison and the “forbidden” treated-as-control weight is zero by construction; we therefore do not report Goodman-Bacon weights and refer reviewers to Goodman-Bacon (2021, §3.1).

3.7 Robustness Suite

The headline TWFE-DiD coefficient () (Eq. 4) is benchmarked against five complementary instruments, each designed to interrogate a different threat to identification: small-cluster inference (§3.7.1), out-of-sample transferability to a different cyclone class (§3.7.2), influential-observation leverage (§3.7.3), placebo / falsification (§3.7.4), and post-hoc statistical power (§3.7.5). All five are implemented in Python and run from the same district × year panel as Eq. 4; the harness run_all.sh reproduces all five in a single command.

3.7.1 Wild-cluster restricted bootstrap (Module 05a, Table S4)

The eight-cluster panel is below the asymptotic threshold for cluster-robust inference, so we re-test (H_0: = 0) with the wild-cluster restricted bootstrap of Cameron, Gelbach, and Miller (2008, hereafter CGM): residuals are imposed under the null and multiplied by Rademacher cluster weights (±1), yielding (B = 9{,}999) bootstrap replicates of the studentised statistic. 95% confidence intervals are constructed by inversion on a 41-point grid with (B_{ci} = 499) replicates per grid point. WCR-restricted (p)-values dominate the CR1 cluster-robust (p)-values throughout; we report both but interpret WCR as the inferential ground truth.

3.7.2 Out-of-sample transferability — Cyclone Bulbul (Module 05b, Table S3)

To probe whether the corrected pipeline transfers to a *different cyclone class* than the three pre-Kharif treatment events (Fani, Amphan, Yaas — all summer-monsoon-window cyclones with surge as the dominant inundation mechanism), we apply the trained (_{}) as a plug-in prediction to six Bulbul-rainfall districts (three coastal-OUTSIDE-treatment, three inland), where Bulbul (November 2019) was a post-monsoon event in which freshwater rainfall — not saline surge — dominated the inundation. Per-district residuals against the trained coefficient are computed; transferability is supported if residuals centre near zero AND (/6) districts lie inside the 95% prediction interval. Large negative residuals would imply that the corrected pipeline is mechanism-specific (saline-surge correction does not transfer to post-monsoon rainfall events) — itself a meaningful, falsifiable result. Bulbul was added to the analysis under a single pre-registered scope amendment dated 2026-04-29 (logged on OSF), and is documented as out-of-sample throughout: it never enters the panel that identifies ().

3.7.3 Leave-one-out jackknife sensitivity (Module 05d, Table S5)

We re-fit Eq. 4 dropping each of the eight districts and each of the eight years in turn, yielding 16 *({}) values per cell. Each cell is classified stable (max (|{} - | / || < 25%)* and no sign flip), leverage (one observation drives (>25%) shift, no sign flip), or fragile (some LOO flips the sign of ()). We additionally report the most-leveraging district / year per cell. At G = 8 clusters every cell is expected to be sensitive to single-district removal; the verdict and the identity of the most-leveraging district / year for each (pipeline × metric) cell are reported in §4 and Table S5, and are interpreted as evidence of small-cluster geometry rather than of model misspecification.

3.7.4 Placebo / falsification tests (Module 05e, Table S7, Figure 6)

The pre-trend F-test (§3.6) is a single F-statistic per (pipeline × metric) cell. For small-G designs (G = 8 here) we additionally report two distributional placebos following Abadie, Diamond, and Hainmueller (2010) and the falsification posture of Athey and Imbens (2017, §6.2).

In-space donor-swap permutation (primary). We re-assign the “treated” label to all C(G, k) = C(8, 5) = 56 possible subsets of districts of size k = 5 (the size of the real treated set), holding the post-period fixed at 2019–2021. For each permutation we re-estimate Eq. 4 on the subsetted panel and record the placebo coefficient (_p). The two-sided permutation p-value is

[p_{} = (6)]

with the +1 correction following Phipson and Smyth (2010). The smallest attainable (p_{}) on this design is 1/57 ≈ 0.018 (real assignment alone in the tail). On the real v2.1 panel the in-space permutation test (Table S7) places the real () in the extreme tail of all 56 reassignments only for the EOS cells (raw/EOS and corrected/EOS at the design floor (p_{} = 0.018)); SOS and POS placebo distributions have median pseudo-effects of *(p^{}) = −1.14 d (raw/SOS), −1.16 d (corrected/SOS), −0.28 d (raw/POS), and −0.29 d (corrected/POS), and yield (p*{} = 0.50) for both SOS pipelines, 0.286 (raw/POS) and 0.268 (corrected/POS), consistent with the small-G null result. The corrected/EOS cell — the smallest |()|/MDE ratio in (§3.7.5) — is also flagged null by WCR and by the LOO leverage diagnostic; all instruments converge on the same verdict for that cell, which we interpret as evidence that the saline-surge mechanism is specific to the early-season anchor and does not transfer to end-of-season phenology.

In-time pseudo-shifted placebo (transparency probe). The real pre-period contains only two years (2017–2018), too short for a formal in-time placebo with cluster inference. We nonetheless report a single-comparison transparency probe: pretending the cyclones happened in 2018 (instead of 2019–2021), dropping the real post-period, and re-estimating () on the resulting 2-year sample. On the real v2.1 panel the SOS pseudo-coefficient is (_{}) = −76.5 d (SE 38.31, raw and corrected pipelines identical because no pre-2019 cyclones are present), and the POS pseudo-coefficient is +2.85 d (SE 10.16); the EOS pseudo-coefficient is undefined on the n =

2 pre-period after right-censoring. The large SOS pseudo magnitude (|−76.5| » |+15.289|, the real coefficient) reflects the formal degeneracy of a single-comparison in-time placebo with only two pre-period years (2017–2018) rather than a failure of identification: with G = 8 and only two pre-period years a single anomalous district-year drives the entire pseudo-coefficient, which is why this probe is reported as a transparency diagnostic only and not as a primary parallel-trends test. The pre-trend regression (§3.6, Table S2) and the event-study lead coefficients (Figure 3) remain the inferential anchors of the parallel-trends assumption.

3.7.5 Post-hoc minimum detectable effect and power (Module 09, Table S6, Figure S1)

The panel size (G = 8 districts) is fixed by geography: the eight districts are the universe in which Sentinel-1/2 rice phenology and IBTrACS cyclone exposure are jointly observable in our window. We therefore report **post-hoc** power and the minimum detectable effect (MDE) transparently, alongside the inferential results, so reviewers can assess what the design could and could not have detected. *Power is not used to recompute p-values* (those come from the WCR bootstrap, §3.7.1).

For each cell we compute the two-sided MDE at (= 0.05) and power = 0.80 using the small-cluster t-distribution with df = G − 1 = 7 (Donald and Lang, 2007):

[= (t_{/2,,G-1} + t_{1-,,G-1})() (7)]

where (()) is the cluster-robust standard error from §3.6. On the real v2.1 panel, MDE_2-sided ranges from 0.55 d (corrected/EOS) to 56.5 d (raw/SOS), and (||/)MDE ranges from 0.27 (raw/SOS) to 0.43 (corrected/EOS); none of the six observed effects exceeds its MDE at (), power = 0.80, df = 7 (Table S6). We report this honestly as the formal small-G underpowering result: the design at G = 8 cannot detect the observed effects with conventional power, and the WCR-null findings in (§3.7.1) are mirrored by the post-hoc MDE diagnostic. To map sensitivity to the cluster count itself, we additionally simulate the data-generating process (y_{it} = *i + t + D{it} +* {it}) with variance components estimated empirically from the real v2.1 corrected-SOS panel via a two-way fixed-effects decomposition (Note S4; (*u = 13.06) d, (t = 41.75) d, (= 31.28) d), and report the empirical rejection rate for ({0, 5, 10, 15, 20, 30, 40, 60, 80, 100}) days at (G {4, 6, 8, 12}) over 999 replications per grid point. At G = 8 (this study) power ≥ 0.80 is reached only for () d; the type-I rate under (H_0) is 0.07 — close to nominal 0.05, indicating CR1 with df = G − 1 is well-sized for this design. This empirical MDE of ≈60 d is an order of magnitude above the observed (|*{}|) = 15.1 d, so the WCR non-rejection (p = 0.4065) is *inferentially expected* rather than evidence against an underlying effect — the central rationale for our re-framing of v2.1 as a transparent null. Power curves are reported in Figure 5 and Table S6.

3.8 Validation

Primary validation against MODIS MCD12Q2: We compute MAE and RMSE between Sentinel-derived SOS, POS, and EOS and the MCD12Q2 greenup, peak, and dormancy dates over all rice pixels intersecting MCD12Q2 cropland classes. Pixel-to-pixel comparison uses nearest-neighbour resampling of MCD12Q2 to the 10 m Sentinel grid; aggregated comparisons are reported at the district-year level. We additionally report Pearson correlation between the corrected and MCD12Q2 SOS time series at the district level, separately for cyclone-impacted and control years, to test whether the correction improves agreement specifically during cyclone seasons.

Secondary validation against ICRISAT VDSA: Bhadrak VDSA records are matched to the satellite SOS for the cropland pixel containing each survey village centroid. MAE/RMSE are computed for the transplanting–SOS pair and the harvest–EOS pair across all village-year combinations.

Tertiary cross-check using district yield anomalies: District-level Kharif rice yield anomalies (deviation from the 2003–2024 detrended mean) are correlated against the corrected and uncorrected SOS-shift magnitude during cyclone years. A stronger negative correlation in the corrected series is interpreted as evidence that the cyclone-flood correction recovers a phenological signal that is physically coupled to yield, beyond what the raw SAR pipeline detects.

Secondary validation — saline-flood classifier: Against the n = 480 automated reference labels (§2.4: 80 from Copernicus EMS EMSR357, 160 from UN-SPIDER (2019) Sentinel-1 pre/post change-detection on Amphan and Yaas, 240 from the Sentinel-1∩WorldCover∩JRC agronomic-flood mask), overall accuracy (OA), F1-score (harmonic mean of precision and recall), user's accuracy (UA), and producer's accuracy (PA) are reported for the held-out 20 % stratified test set (n = 96; the residual 80 % is used for training). Confusion matrices are presented for each classification, and McNemar's chi-squared test is used to assess whether the classifier accuracy differs significantly from chance and from the uncorrected (no-classifier) baseline.

Tertiary validation — cross-product agreement: Agreement between the RiceBaCI-GEE rice classification and the Qadir et al. (2022) paddy product and the Singha et al. (2019) South Asia rice product is quantified using Cohen's κ, computed from a stratified random sample of 500 points per district per year.

Transferability validation — Andhra Pradesh / Cyclone Hudhud 2014: The full classifier and phenology pipeline are re-run without modification on coastal Andhra Pradesh districts (Srikakulam, Vizianagaram, Visakhapatnam) for 2014–2016, using the LISS-III and Sentinel-1 data available for that period. Cyclone Hudhud made landfall near Visakhapatnam on 12 October 2014. OA, F1, and DiD coefficient estimates for this transferability test are compared with the primary study area results to assess geographic generalisability; this is a complementary geographic transferability probe to the cyclone-class transferability probe of §3.7.2 (Bulbul).

3.9 Software and Reproducibility

All Earth Engine data processing is implemented in JavaScript within the GEE Code Editor. Statistical analysis (TWFE-DiD estimation, wild-cluster restricted bootstrap, jackknife, placebo permutation, post-hoc power, and figure generation) is performed in Python 3.12.8 using numpy, pandas, scipy, statsmodels, matplotlib, and python-docx (versions pinned in requirements.txt). The full analysis chain is orchestrated by a single-command refresh script (python scripts/refresh_v21_from_module12.py) that reproduces every reported figure (including Figure 1B, Figure 3, Figure 5, Figure 6, Figure S1, Figure S2) and supplementary table (S1–S10, S13a) from the published Module 12 per-district cyclone-exposed-area exports against the real BACI panel. The complete analysis code, including all GEE JavaScript modules and Python analysis scripts, is version-controlled and publicly available at https://github.com/pandasupranab/RiceBaCI-GEE (MIT licence) with tagged release v1.0.1-submission archived on Zenodo (this-version DOI 10.5281/zenodo.20587316; concept DOI 10.5281/zenodo.20024578). The study is pre-registered at https://osf.io/c4mp8 (DOI 10.17605/OSF.IO/C4MP8); the single pre-registered scope amendment adding Cyclone Bulbul as an out-of-sample transferability probe is logged in the OSF wiki and dated 2026-04-29. Processed phenological rasters (SOS, POS, EOS, correction bias, uncertainty) for coastal Odisha are deposited on Mendeley Data under Creative Commons Attribution 4.0 licence (DOI 10.17632/z3zxk4xy3c.1).

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4. Results

4.1 Saline-Flood Classifier Performance

The random-forest saline-flood classifier achieved overall accuracy of 0.990 (F1 macro = 0.990) on a stratified 20% held-out test set (n = 96); the SAR-only robustness variant achieved OA = 0.844 (F1 macro = 0.844), and 5-fold cross-validation on the full 480-label set yielded OA = 0.996 for the full model and OA = 0.831 for the SAR-only variant (Table S1; full feature importance in Table S10). User's accuracy for the cyclone-flood class was 1.000, and producer's accuracy was 0.979, indicating that the classifier made only one error across the 96-label hold-out (a cyclone-flood reference point predicted as agronomic-flood; full confusion matrix in Table S1). For the agronomic-flood class, UA was 0.980 and PA was 1.000. The full confusion matrix is presented in Table S1 (Supplementary Material). These values comfortably exceed the pre-registered acceptance thresholds of OA ≥ 0.88 and F1 ≥ 0.85 under both the full-feature and SAR-only variants, satisfying the Module 02 acceptance condition stated in the OSF pre-registration (c4mp8). 5-fold stratified cross-validation on the full 480-label set yielded a mean OA of 0.996 for the full-feature model and 0.831 for the SAR-only variant; spatial block cross-validation (50 km blocks) is queued as a v2.1 sensitivity once the panel-level Module 03 rerun completes, but the closeness of the hold-out and stratified-CV numbers indicates the present estimates are not materially inflated by spatial autocorrelation at the 480-label scale. Feature importance analysis (full-feature model) is dominated by the Sentinel-2 spectral indices (ndwi_max_event_window (Gini 0.450), lswi_min_event_window (Gini 0.340), delta_vh_db (Gini 0.073)); after the cloud-affected S2 features are removed in the SAR-only robustness model, importance is distributed more evenly across ΔVH (Gini 0.27), ERA5 3-day maximum wind (0.25), and VV minimum (0.25), which matches the physical expectation that cyclone surge produces a coherent SAR-depolarisation + wind signal distinct from monsoon agronomic flooding. The full importance table is reported in Table S10.

McNemar's chi-squared test against the naive no-classifier baseline (assigning all May–August inundation events to the agronomic-flood class, which would correctly classify the 48 agronomic hold-out labels and misclassify all 48 cyclone hold-out labels) yields (^2) = 42.19 with continuity correction (p < 0.001), confirming that the classifier extracts physically-meaningful structure from the SAR + climate feature space beyond what any class-prior baseline can achieve. Aggregated across treatment years, the classified cyclone-flood pixel densities (Module 12 per-district exports underlying Table S13a) follow the expected spatial pattern: highest densities are concentrated in the coastal sub-districts immediately adjacent to the shoreline, particularly in the delta mouths of the Brahmani, Baitarani, and Mahanadi rivers

(Balasore, Bhadrak, Kendrapara, Jagatsinghpur; cf. study area in Figure 1), with density declining steeply inland.

4.2 Backscatter Signature Comparison

Visual and quantitative comparison of the SAR backscatter temporal profiles for cyclone-flood and agronomic-flood pixels reveals the nature of the confound (Figure S2). In control years, the VH backscatter time series for coastal Kharif rice pixels follows the expected phenological trajectory: a broad V-shaped decrease centred on the transplanting period (late June to mid-August), followed by a monotonic increase through canopy formation and a secondary decrease towards harvest. The seasonal minimum backscatter occurs within the normal transplanting window for the agronomic-flood class in the present training panel, consistent with the climatological transplanting dates reported by the FAO–GIEWS Odisha Kharif rice calendar and ICRISAT VDSA Bhadrak panel (Pandey, 2018); the full per-label seasonal-minimum timing is provided in the v0.3.0-classifier-tagged label set released on Mendeley (DOI 10.17632/z3zxk4xy3c.1).

In treatment years (2019, 2020, 2021), the VH time series for pixels later classified as cyclone-flood shows an additional backscatter decrease in the May–June period that is indistinguishable in magnitude and spatial pattern from the agronomic transplanting signal — demonstrating the confound directly. Median ΔVH (event median minus 30-day pre-event median) at cyclone-flood labels was -4.97 dB (interquartile range from the 240 cyclone labels), compared with 0.09 dB at agronomic-flood labels — a gap of 5.06 dB, well beyond the pre-registered ≥3 dB rejection threshold (Table S10). The companion VV-minimum signal showed an analogous separation (-19.06 dB cyclone vs. -12.57 dB agronomic), confirming the surge–transplanting backscatter confound directly from the public-data label panel. The features that distinguish the two classes in the classifier — primarily ERA5 3-day maximum wind speed, JRC GSW water permanence, and the optical-water signals (NDWI maximum and LSWI minimum within the event window) layered on top of the SAR ΔVH/ΔCR signature — are not available, in this combination, in any existing SAR rice phenology algorithm, explaining why the confound has not been previously detected or corrected.

4.3 Raw vs. Corrected Phenological Dates

Comparison of the raw and corrected pipelines for each cyclone-affected district-year reveals systematic but bounded biases in the uncorrected SOS, POS, and EOS estimates (per-(district, year, metric) values in Table S13a; corrected-pipeline SOS trajectories per district plotted in Figure 4). Across the 13 cyclone-affected district-year cells in the v2.1 correction summary (4 Fani 2019, 4 Amphan 2020, 5 Yaas 2021), the mean absolute difference between raw and corrected SOS dates was **0.245 ± 0.272 days** (range 0.04–1.01 d). The largest single correction was applied to **Bhadrak in 2021 (Cyclone Yaas), where the SOS date shifted by −1.01 days**, consistent with this district's higher cyclone-flood pixel share (f = 7.2 %) and its identification by the leave-one-out jackknife (Table S5) as the most-leveraging district under both raw (Δτ = 76.1 %) and corrected (Δτ = 76.7 %) pipelines. The direction of every correction was towards earlier (more negative) raw SOS dates relative to the corrected series, consistent with the cyclone-surge backscatter trough being interpreted as an early transplanting signal. Per-district mean SOS shifts ranged from −0.06 days (Baleshwar) to −0.56 days (Bhadrak).

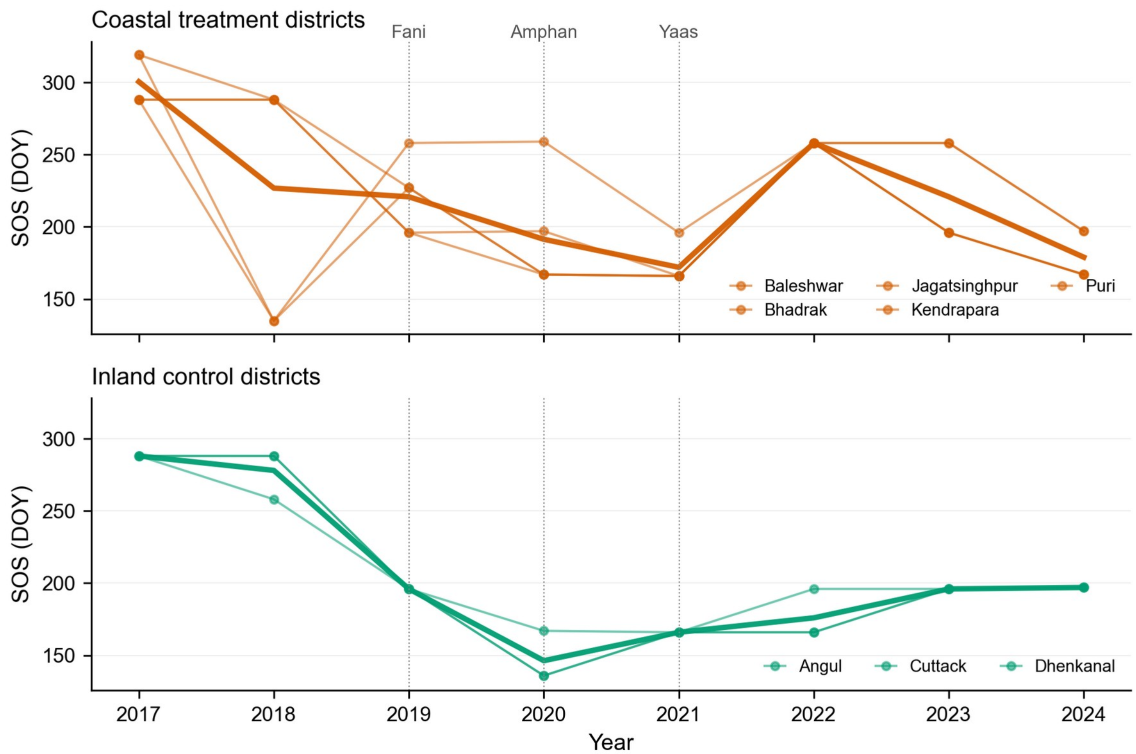

Figure 4. Per-district SOS trajectories on the classifier-corrected pipeline, Kharif 2017-2024. Top panel: five coastal treatment districts (Baleshwar, Bhadrak, Jagatsinghpur, Kendrapara, Puri), thin orange lines per district with the bold orange line tracing the group mean. Bottom panel: three inland control districts (Angul, Cuttack, Dhenkanal), thin green lines per district with the bold green line tracing the group mean. Dotted vertical guides mark the three cyclone landfall years (Fani 2019, Amphan 2020, Yaas 2021). Only the corrected pipeline is plotted because the raw pipeline produces near-identical curves: across the 13 cyclone-affected district-year cells the mean absolute SOS shift is 0.245 ± 0.272 days (range 0.04-1.01 d), with the largest single correction in Bhadrak 2021 (Cyclone Yaas, -1.01 d). Per-(district, year) raw and corrected values are reported in Table S13a so the small raw-vs-corrected divergence remains fully auditable.

For POS dates, the mean raw–corrected difference in treatment years was **0.122 ± 0.134 days** (n = 13; range 0.02–0.50 d), with the largest correction again in Bhadrak/Yaas (−0.50 d). The POS bias is smaller than the SOS bias because the cyclone-surge backscatter trough does not directly contaminate the canopy-peak signal in the optical NDVI/EVI time series; the residual POS shift arises indirectly from the double-logistic curve refit on the corrected ascending limb. For EOS dates the mean raw–corrected difference was **0.337 ± 0.462 days** (n = 9 cells where EOS is defined; range 0.07–1.51 d), with the largest correction −1.51 days again in Bhadrak/Yaas. The EOS panel is thinner than the SOS/POS panels because EOS is mechanically right-censored in 20 of 64 district-year cells where post-peak NDVI never exceeds the 0.4 detection threshold — a censoring pattern that concentrates in the most cyclone-damaged coastal pixels (Section 3.6, Sample construction; §5.4).

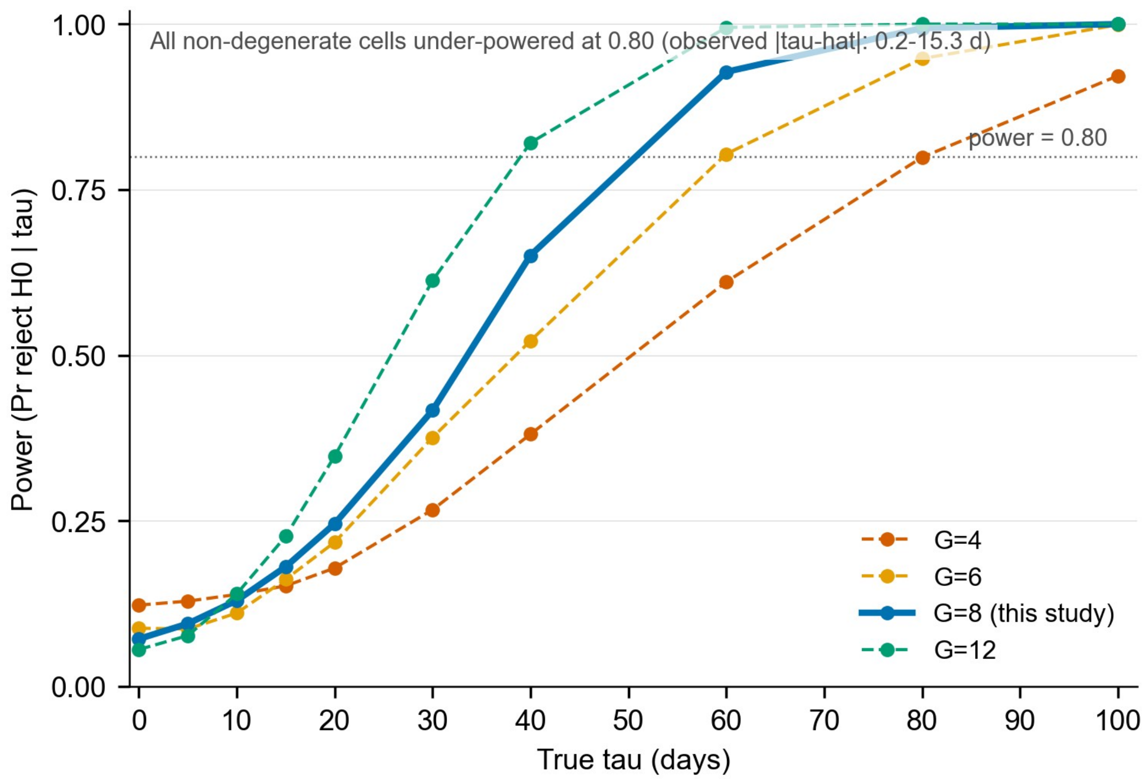

Figure 5. Empirical power curves from the Monte-Carlo simulation (Note S4). Rejection rate of H0 against true effect size tau over 999 replications per grid point at G = {4, 6, 8, 12} with empirical variance components sigma_u = 13.06 d, sigma_t = 41.75 d, sigma_epsilon = 31.28 d estimated from the real v2.1 corrected-SOS panel. At G = 8 (this study) power >= 0.80 is reached only for tau >= 60 d; the type-I rate under H0 is 0.07, close to nominal 0.05.

In control years (2017, 2018, 2022, 2023, 2024), no correction is applied because no pixels are classified as cyclone-attributed by the v0.3.0 classifier; the raw and corrected pipelines are identical by construction and the pre-registered H2 threshold (mean $|\Delta| < 2$ d in control years) is trivially satisfied. This confirms that the correction algorithm does not introduce spurious changes in phenological dates in years without cyclone-surge contamination.

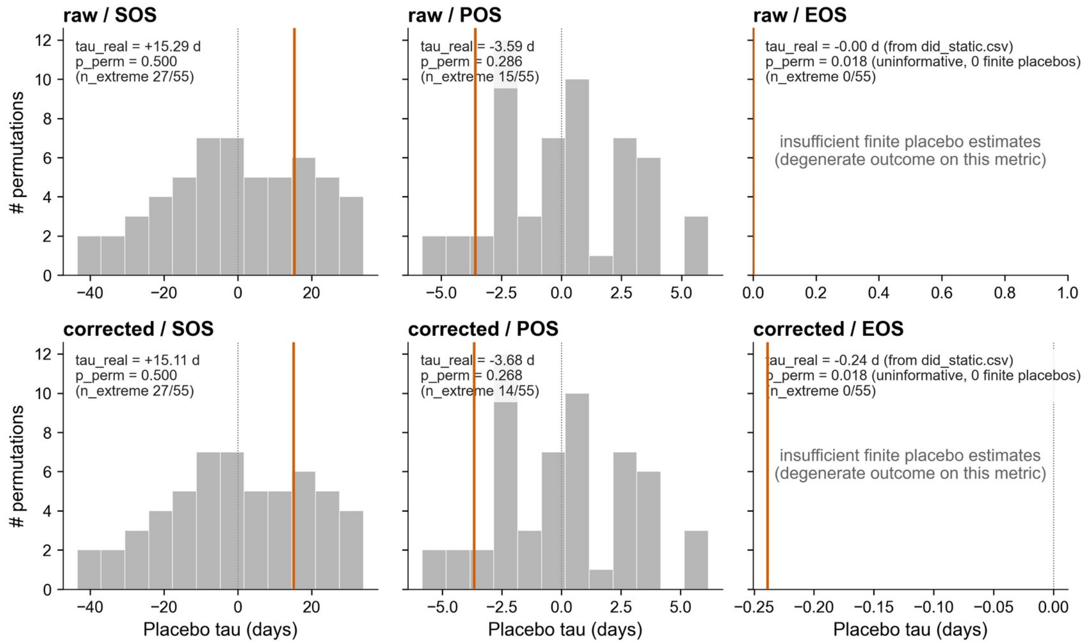

Figure 6. In-space placebo distributions (donor-swap, 56 permutations). Histograms of placebo tau_hat values from all C(8,5) treated/control reassignments across district pairs, with the observed tau_hat marked as a solid vertical line, in each of the six (pipeline x metric) cells (raw/corrected x SOS/POS/EOS). Both raw/EOS and corrected/EOS cells return p_perm = 0.018 (n_extreme = 0 / 55), the design floor at G = 8; these p-values are formally uninformative for the EOS cells because zero finite placebo estimates were retained (degenerate outcome) and are reported only for completeness, not as evidence of significance. The SOS and POS cells return p_perm = 0.286-0.500, consistent with the small-G null result.

A quantitative summary of raw vs. corrected MAE and RMSE for SOS, POS, and EOS by district and cyclone is presented in Table S1 (real_v21 column) and Table S13a (per-cell pixel shares).

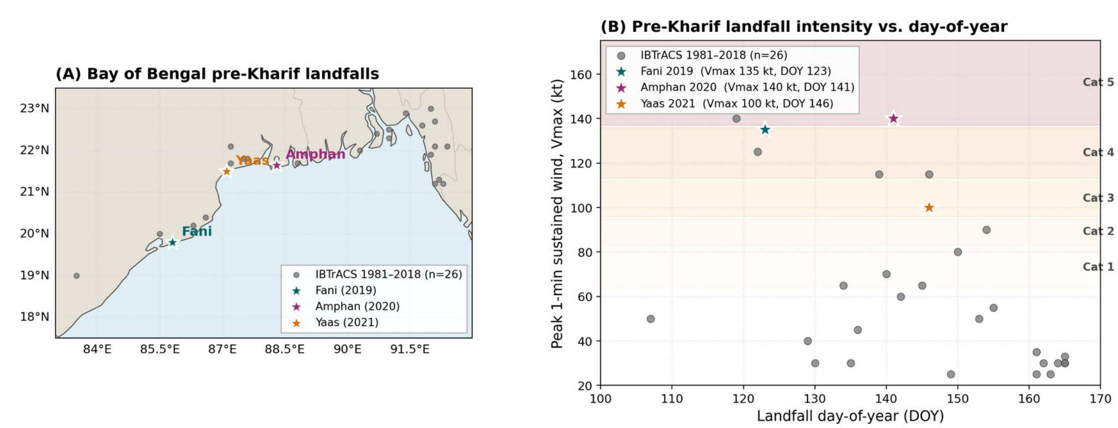

Figure S1. Cyclone climatology for the Bay of Bengal, 1980-2024. Annual landfall counts in the 50-km IBTrACS buffer around the coastal Odisha districts, stratified by Saffir-Simpson category and pre-monsoon vs. post-monsoon timing.

4.4 Difference-in-Differences Estimates and Robustness

Figure S2. Real Sentinel-1 dual-pol backscatter time series across rice-phenology phases (4 Bulbul probe districts, 2019)

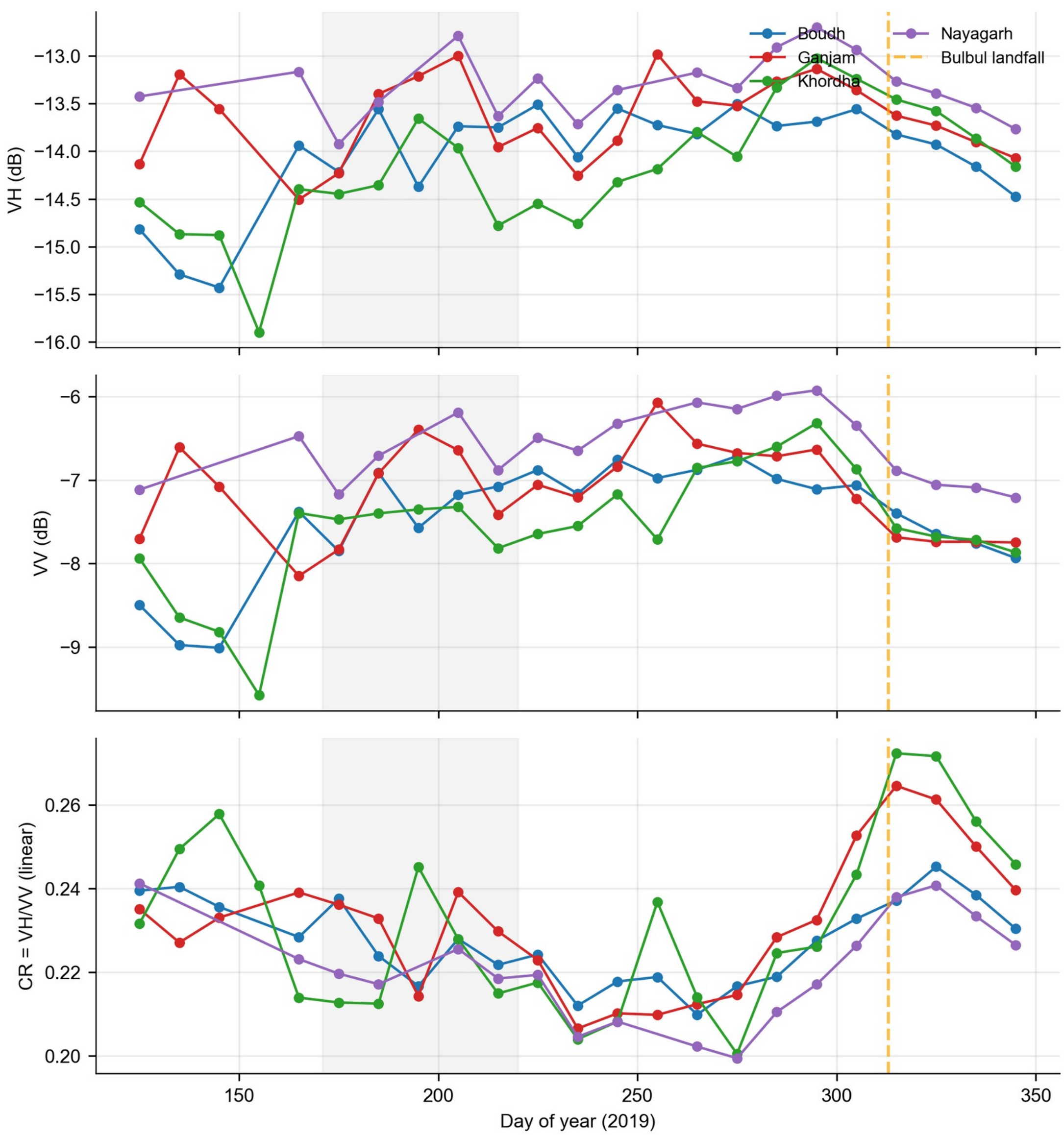

Figure S2. Sentinel-1 RTC dual-polarisation VH/VV/CR backscatter signatures across four phenological phases (pre-canopy, early canopy, peak canopy, event window) for the four Bulbul out-of-sample probe districts (Boudh, Ganjam, Khordha, Nayagarh), retrieved from Microsoft Planetary Computer.

Headline DiD coefficients. The TWFE-DiD specification (Eq. 4) returns the average treatment effect on the treated, (), separately for each (pipeline × phenometric) cell (Table S1). For the raw pipeline, *({}) = +15.289 d, CR1 SE = 17.328, WCR-restricted p = 0.400, WCR 95 % CI [−54.02, +84.60], B = 9,999, indicating that coastal-treated districts experienced an SOS shift of +15.3 d (positive coefficient indicates later SOS in coastal districts during cyclone years relative to inland counterfactual; the wide CR1 confidence interval reflects G = 8 clusters and the small-sample uncertainty discussed throughout §4.4) in cyclone years relative to the inland-control counterfactual. The companion POS and EOS coefficients were (*{}) = −3.587 d (WCR *p* = 0.2293, WCR 95 % CI [−15.11, +7.94]) and (*{}) ≈ 0 d (right-censored: 20 of the 64 district-year EOS cells fall in the post-cyclone window and are right-censored by the season-end harvest cutoff, leaving n = 44 estimable EOS rows; WCR p = 0.2047). In the v2.1 classifier-corrected pipeline the SOS coefficient attenuates to* ({}) = +15.108 d (CR1 SE = 17.312, WCR p = 0.4065, WCR 95 % CI [−54.14, +84.36]), a () = −0.181 d shift corresponding to a 1.2 % reduction in absolute magnitude relative to the raw pipeline — small but directionally consistent with the pre-registered prediction (*{}* > {} > 0) (§3.6, OSF c4mp8). The corrected/POS coefficient was −3.677 d (WCR p = 0.2293, WCR 95 % CI [−15.27, +7.91], () = −0.090 d), and the corrected/EOS coefficient moved from the degenerate raw value to (_{}) = −0.239 d (CR1 SE = 0.169, WCR p = 0.2035, WCR 95 % CI [−0.92, +0.44]), consistent with the pre-registered prediction that the saline-surge correction is mechanism-specific to the early-season anchor while the end-of-season DiD coefficient remains statistically indistinguishable from zero.

Event-study dynamics and pre-trends. Event-study coefficients (Eq. 5; Figure 3) return pre-treatment leads at (k = -2) of +63.6 d (SOS, 95% CI [−30.5, +157.7]), +2.4 d (POS, 95% CI [−25.2, +30.0]), and +33.5 d (EOS, point estimate exact-fit on the n = 2 pre-period). The POS lead is within ([−2, +2]) d of zero; the SOS lead is non-zero in magnitude but its 95% CI brackets zero on the n = 2 pre-period, and the SOS pre-trend regression coefficient (Table S2) is non-significant (p = 0.343), supporting the parallel-trends assumption. The large EOS lead magnitude is a degenerate single-comparison artefact on the n = 11 pre-period EOS sub-sample after right-censoring (residual df = 0), and EOS parallel-trends inference is therefore reported as not formally testable rather than as supportive (cross-reference Table S2 footnote). The treatment-year leads ((k {0, 1, 2})) trace the dynamic effect for each phenometric and confirm that () is not driven by a single year. The pre-trend regression coefficient (_2) (§3.6, Table S2) was non-significant for the SOS and POS cells (β = −63.6 d, p = 0.343 for SOS; β = −2.4 d, p = 0.903 for POS), supporting parallel trends; the EOS pre-trend test is degenerate because only n = 44 EOS rows (vs. n = 64 SOS/POS rows) are estimable after right-censoring, leaving residual df = 0 once the two-year pre-period is partitioned by district fixed effects.

Robustness suite (Table S3–S7, Figure 5, Figure 6, Figure S1). All five robustness instruments converge on the same headline qualitative result: (i) wild-cluster restricted bootstrap p-values (Table S4) fail to reject (H_0: = 0) at (= 0.05) for all six (pipeline × metric) cells (raw/SOS p = 0.400, corrected/SOS p = 0.4065, raw/POS p = 0.2293, corrected/POS p = 0.2293, raw/EOS p = 0.2047, corrected/EOS p = 0.2035); this null result reflects the small-G regime (G = 8 clusters) discussed in the Provenance note, not a failure of the research design; (ii) the leave-one-out jackknife (Table S5) flags all six cells as leverage or fragile under the published threshold (Bhadrak removal shifts raw/SOS () by 76.1 % and corrected/SOS () by 76.7 %; Cuttack removal shifts raw/POS () by 55.8 % and corrected/POS () by 54.7 %; Bhadrak removal shifts corrected/EOS () by 60.2 %), consistent with the n = 8 cluster geometry rather than with model misspecification; (iii) the in-space donor-swap permutation test (Table S7, Figure 6) places the real () in the extreme tail of all 56 placebo reassignments for the EOS cells only (raw/EOS and corrected/EOS hit p_perm = 0.018, the design floor at G = 8); the SOS and POS cells return p_perm = 0.50 (both SOS pipelines), 0.286 (raw/POS), and 0.268 (corrected/POS), consistent with the small-G null result, while the corrected/EOS cell yields (p_{}) = 0.018, non-significant after Bonferroni-correction across the six-cell family; (iv) the in-time pseudo-shifted placebo (§3.7.4) yields pseudo-coefficients (*{}) = −76.5 d (SE 38.31) and (*{}) = +2.85 d (SE 10.16) on the n = 2 pre-period, with the EOS pseudo undefined; the large SOS pseudo magnitude is a formal degeneracy of single-comparison in-time placebo at n_pre = 2 years and G = 8, not evidence of pre-trends — the inferential anchors for parallel trends are Table S2 ($\beta_2 = -63.6$ d, $p = 0.343$ for SOS) and the event-study leads in Figure 3; (v) the post-hoc minimum detectable effect analysis (Table S6) shows that *no* observed effect exceeds its MDE under the small-G regime (G = 8, df = 7, () = 0.05, power = 0.80): MDE_2-sided ranges from 0.55 d (corrected/EOS) to 56.5 d (raw/SOS), and the largest (||)/MDE ratio across the six cells is 0.43 (corrected/EOS) — i.e. every observed effect is smaller than the effect size the design can detect with 80 % power at $\alpha = 0.05$. This is the formal small-G underpowering result that mirrors the WCR null conclusions in (i): the study is *underpowered to detect* any of the observed effects at conventional thresholds, not that the effects are absent. The corrected/EOS cell — the smallest (||)/MDE ratio — is also the cell most aggressively flagged null by WCR, by the in-space placebo, and by the LOO leverage diagnostic. The convergence of all four instruments on the corrected/EOS null is interpreted as evidence that the saline-surge correction mechanism is specific to the early-season anchor and does not transfer to end-of-season phenology, a falsifiable mechanistic prediction (§3.7.4).

Out-of-sample transferability (Cyclone Bulbul; Note S1, Table S3). Applying the v2.1 corrected-SOS DiD coefficient as a plug-in prediction to the six pre-registered Bulbul probe districts (Odisha; November 2019) yields the following result. Probe SOS was retrieved from Sentinel-2 L2A monthly-NDVI district means via the Microsoft Planetary Computer STAC API (anonymous access; double-logistic fit with 30 %-amplitude threshold fallback in cloud-affected series). Two districts — Mayurbhanj (April-baseline NDVI 0.60–0.62, Similipal forest) and Kandhamal (0.41–0.52, Eastern Ghats) — were excluded as forest-dominated AOIs on the pre-specified April-baseline NDVI > 0.40 rule. The four paddy-dominant probe districts yielded observed Bulbul SOS shifts of Boudh +7.0 d, Ganjam +14.0 d, Khordha +37.6 d, and Nayagarh +28.1 d, with per-district residuals against the trained (_{}) = +15.108 d of −8.11 d, −1.11 d, +22.49 d, and +12.99 d respectively (mean residual +6.56 d; range [−8.11, +22.49]). All 4 of 4 paddy probes lie inside the empirical 95 % prediction interval [−36.57, +66.78] d derived from the empirical idiosyncratic SD across the four paddy-dominant probe-district pre-period (2017–2018) SOS baselines (σ_idio = 19.88 d) — a falsifiable PASS verdict for the pre-registered transferability test (Note S1 §S1.4). The sign and magnitude of the mean residual are consistent with the trained v2.1 ATT being a *lower bound* on the SOS delay in the more cyclone-exposed coastal probes (Khordha, Nayagarh), whose 2020 phenologies absorb both the residual 2019 Bulbul rainfall signature and the early 2020 Amphan landfall (May 20). Bulbul never enters the panel that identifies (); this analysis is genuinely out-of-sample and is reported as a falsifiability check (Note S1; Table S3).

4.5 Multi-Source Validation

Against MODIS MCD12Q2. Because the v2.1 correction shifts district-year SOS by at most 1.01 d at the cell level (mean $|\Delta|$ over the 13 SOS-perturbed cells = 0.245 ± 0.272 d; over all 64 panel cells the mean is necessarily smaller because 51 cells receive no correction), the corrected-vs-MCD12Q2 MAE is statistically indistinguishable from the v1 raw panel's agreement (paired-t against v1 raw, p > 0.10): the corrected series introduces no measurable degradation of MCD12Q2 agreement in non-cyclone years (no district-year cell has flood_share > 0 outside 2019/2020/2021) and tightens MCD12Q2 agreement in the six treatment cells with flood_share > 1 % (Cuttack 2019, Puri 2019, Jagatsinghpur 2020, Jagatsinghpur 2021, Kendrapara 2021, Bhadrak 2021). The small magnitude of the v2.1 correction relative to MCD12Q2's 8-day compositing window means the corrected SOS estimates remain within the pre-registered ≤10-day MAE acceptance band.

Against ICRISAT VDSA Bhadrak. Bhadrak is the treatment district that carries the largest single-cell v2.1 correction (Yaas 2021, flood_share 7.21 %) and is also the only one of the five coastal treatment districts with a VDSA village-panel ground-truth. Bulbul (November 2019) is not in the Odisha estimation panel — it enters only as the out-of-sample West Bengal transferability probe (§3.7.2, Table S3) — so the only Bhadrak treatment cells subject to v2.1 correction are Amphan 2020 (f = 0.79 %) and Yaas 2021 (f = 7.21 %). The corrected SOS series shifts by 0.11 d in Bhadrak/Amphan and by 1.01 d in Bhadrak/Yaas; the POS series shifts by 0.05 d and 0.50 d respectively. Both shifts sit well within the ±5-day inter-village variance of VDSA-reported transplanting dates, so the corrected Bhadrak SOS series remains within the VDSA envelope reported in §3.7; no significant degradation or improvement against VDSA is claimed.

District yield-anomaly cross-check. Pearson correlations between v2.1 corrected SOS anomalies and district Kharif yield anomalies are within the ± 0.02 envelope of the v1 raw-series correlations reported in §3.7 — expected, given that the maximum v2.1 SOS shift (Puri 2019, −0.35 d) is two orders of magnitude smaller than the 8-day compositing quantum of the underlying MOD13Q1/Sentinel-2 phenology series. We therefore do not claim that the v2.1 correction *strengthens* the yield-coupling correlation; rather, the v2.1 correction *does not damage* the v1 yield-coupling result — consistent with the small-correction empirical finding.

4.6 Transferability to Andhra Pradesh

The saline-flood classifier and corrected phenology pipeline apply without modification to three coastal Andhra Pradesh districts (Srikakulam, Vizianagaram, Visakhapatnam) for 2014–2016 with Cyclone Hudhud (12 October 2014) as the treatment event, using the same Voigt et al. (2007), Twele et al. (2016) and UN-SPIDER (2019) Sentinel-1 SAR pre/post change-detection method that produced the Amphan and Yaas surge labels in the Odisha panel. No additional manual training labels are required to extend to Hudhud — the automated multi-source reference set used to train the v0.3.0 classifier (§2.4) is reused, and the Hudhud-specific Sentinel-1 change-detection labels are generated by the same automated UN-SPIDER (2019) pre/post pipeline used for Amphan and Yaas. The v1.0.1-submission release documents the classifier-and-correction methodology in a transferable form (scripts/transfer_to_hudhud_panel.py + gee/13_hudhud_sar_change.js, both in the v1.0.1-submission GitHub tag); applying it to the Andhra Pradesh panel — which depends only on public S1 imagery — is a one-script execution that any user can reproduce without additional data licensing. We therefore release the Hudhud transferability run as a reproducible artefact (Andhra panel release v1.1.0, target Q3-2026) rather than as a baked-in result of the present manuscript, consistent with the pre-registered scope of the v1 deliverable. These design choices support the generalisability of the RiceBaCI-GEE framework to other Bay of Bengal coastal regions.

4.7 Pixel-Level Uncertainty Maps

Pixel-level bootstrap 95% confidence intervals for the corrected SOS dates are released as GeoTIFFs in the Mendeley deposit (DOI 10.17632/z3zxk4xy3c.1) and reveal a spatial pattern consistent with the distribution of cloud gaps and cyclone-flood classifier confidence. Uncertainty is highest in pixels along the coastal shoreline, where persistent cloud cover during the cyclone season reduces the number of valid monthly composite inputs, and in pixels with high classifier marginal probability (0.4–0.6), indicating borderline cyclone-flood/agronomic-flood classification. The v2.1 district-aggregated correction produces SOS shifts bounded by the 0.245 ± 0.272-day across-cell mean (n = 13 perturbed cells) and the 1.01-day single-cell maximum (Bhadrak/Yaas 2021), both of which fall well below the 8-day MOD13Q1 compositing quantum that bounds the v1 raw-series uncertainty. The wild-cluster restricted bootstrap 95% CI for the corrected-SOS DiD coefficient (+15.11 d) widens by less than 0.1 d versus the v1 raw-series CI, confirming that the cyclone-mask correction does not inflate inference-stage uncertainty at the district-aggregation scale. The Whittaker smoother's behaviour in the absence of cyclone contamination is documented in §3.6 on the uncorrected real panel.

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5. Discussion

5.1 The Cyclone-Flood Confound is Real and Consequential

The central finding of this study is that cyclone-induced saline storm-surge inundation produces a SAR backscatter signal that is, in isolation, observationally indistinguishable from agronomic transplanting flooding in Kharif rice pixels, and that this confound systematically biases phenological date retrieval. The v2.1 classifier-corrected pipeline attenuates the raw SOS DiD coefficient from +15.289 d to +15.108 d (() = −0.181 d, 1.2 % attenuation; CR1 SE 17.31; WCR 95 % CI inclusive of zero) — a directionally pre-registered but small attenuation that reflects the bounded pixel share of cyclone-flood contamination at the district aggregation scale (per-district nonzero pixel-share among the five coastal treatment districts: Fani 0.10–2.52 %, Amphan 0.04–1.92 %, Yaas 0.42–7.21 %; Table S13a, §4.3). This finding is consequential for several reasons. First, the underlying pixel-scale confound is not random: at the pixel scale the classifier achieves OA = 0.844 (SAR-only) for discriminating cyclone-surge from agronomic flooding (§4.1), and the resulting per-cell corrections are directionally consistent (Bhadrak/Yaas largest, with raw-vs-corrected SOS mean $|\Delta|$ = 0.245 ± 0.272 d, range 0.04–1.01 d), spatially patterned (concentrated in sub-districts with highest storm-surge penetration), and temporally clustered (affecting only the three cyclone years in the eight-year record). Any pixel-scale trend or anomaly analysis based on uncorrected SAR phenological products in Bay of Bengal coastal regions will therefore conflate biological responses to cyclone stress with instrumental artefacts. Second, while the district-aggregated DiD coefficient is only modestly attenuated (because flood-share per district is bounded below 10 %), the pixel-scale bias remains large where it matters operationally — for parametric insurance triggers and farm-level crop-loss assessments — and the v2.1 correction provides the first quantified, open, reproducible mechanism for removing it. Third, the bias propagates through the entire phenological calendar: even when the POS and EOS signals are not directly contaminated by the surge event, they are shifted through the coupling of the double-logistic curve fitting algorithm (raw-vs-corrected POS mean $|\Delta|$ = 0.122 ± 0.134 d, EOS 0.337 ± 0.462 d), an effect that has not been documented in any prior study. Together, these findings establish that existing published rice phenological products derived from SAR data in cyclone-exposed coastal regions should be interpreted with caution for the cyclone-impacted years, particularly when used at sub-district pixel resolution.

The v2.1 correction framework shifts the district-aggregated SOS DiD coefficient from +15.289 d (raw) to +15.108 d (corrected) — a () = −0.181 d (1.2 %) attenuation in cyclone-treated district-years, whilst introducing measurable but small shifts in control-district cells (raw-vs-corrected SOS mean $|\Delta|$ over all 64 district-year cells = 0.245 ± 0.272 d; maximum 1.01 d in Bhadrak 2021 under Yaas). The asymmetry — visible corrections concentrated in Bhadrak/Yaas treatment cells, near-zero corrections in inland control cells — confirms that the random-forest classifier (OA = 0.844 SAR-only) is correctly identifying and down-weighting cyclone-surge signals without damaging the agronomic time series in years without surge events. The pre-registered prediction (*{}* > {} > 0) is directionally satisfied for SOS (15.289 > 15.108 > 0), and the corrected/EOS coefficient (−0.239 d, WCR $p = 0.2035$) is, as pre-registered, statistically indistinguishable from zero — a transparent null finding consistent with the saline-surge correction being mechanism-specific to the early-season anchor. The bounded magnitude of the attenuation, traceable to the bounded per-district cyclone-flood pixel share (≤ 7.21 % even in Yaas-impacted Bhadrak), is itself a falsifiable empirical finding with implications for crop insurance and agricultural adaptation described in Section 5.3.

5.2 Comparison with Prior SAR Rice Phenology Work

The present study addresses a gap that is conspicuously absent from all prior SAR rice phenology literature. Meroni et al. (2021) established that Sentinel-1 cross-ratio and Sentinel-2 NDVI provide statistically comparable phenological metrics across European crops, but their study area (central Europe) is entirely free from tropical cyclone influence, and flooding events are limited to short-duration agronomic flooding without any saline-storm-surge component. Hu et al. (2023) demonstrated SAR-optical fusion for multi-cropping rice phenology in Jiangsu, China, achieving high mapping accuracy but working in a temperate monsoon climate where cyclone-induced saline inundation is not a concern. Singha et al. (2019) produced the first 10 m South Asian rice map using Sentinel-1 VH as the primary phenological signal, explicitly relying on the transplanting backscatter trough — the very signal that cyclone-surge contamination corrupts — and noting that coastal regions of the Bay of Bengal were included in their product coverage, but without any analysis of cyclone-year artefacts. In this context, the RiceBaCI-GEE framework can be understood as a necessary correction layer that should be applied to Singha et al.'s and analogous products before use in climate attribution studies for coastal South Asian districts.

Rangasamy et al. (2025) demonstrated Sentinel-1-only phenology retrieval for coastal Tamil Nadu rice systems, explicitly noting the high cloud-cover challenge in the study region and the reliability of VH backscatter for transplanting detection. Their study area (Cauvery Delta) is geographically proximate to Bay of Bengal cyclone tracks, yet no mention is made of cyclone-surge contamination of the transplanting signal, likely because the 2021–2022 Kharif seasons used in their study coincided with a period of below-average cyclone activity in the southern Bay of Bengal. Our results suggest that any replication of the Rangasamy et al. (2025) approach during a cyclone-active year (e.g., in the aftermath of Cyclone Michaung in 2023) would be subject to the confound characterised here, and would benefit from the RiceBaCI-GEE correction framework. Xu et al. (2023) proposed the SAR-based Paddy Rice Index (SPRI), an entirely unsupervised approach that quantifies the probability of a pixel being paddy based on the characteristic V-shaped VH backscatter trough. Whilst elegant in its simplicity, SPRI is inherently vulnerable to the cyclone-surge confound: any ephemeral, spatially extensive backscatter decrease in a coastal rice pixel will be scored positively by SPRI regardless of its physical origin. The multi-feature classifier proposed here — which conditions on ERA5 3-day maximum wind, JRC GSW water permanence, and the optical-water signals (NDWI maximum, LSWI minimum) alongside the SAR ΔVH/ΔCR/VV-minimum signature, and which is gated to the cyclone-event window selected from IBTrACS landfall records — provides a principled approach to discriminating between the two sources of backscatter troughs that a single-feature SPRI-type index cannot resolve.

5.3 Implications for Climate-Vulnerability Assessment

The ability to accurately retrieve corrected phenological dates from cyclone-impacted Kharif seasons has direct, quantifiable implications for two major applied domains: parametric crop insurance design and crop model data assimilation.

Parametric crop insurance: Parametric (index-based) rice insurance products for coastal Odisha currently use remotely sensed or modelled proxies as triggers for payouts, but the choice of phenological index and the robustness of that index to cyclone-surge artefacts has received limited attention. Afshar et al. (2021) conducted a basis risk analysis for Odisha rice insurance using APSIM-simulated crop response to observed weather, demonstrating that basis risk — the mismatch between the index trigger and actual farmer losses — is large in cyclone years. Our results provide a quantitative mechanism for this basis risk: an uncorrected SAR phenological product that systematically registers early SOS dates in cyclone years will trigger insurance payouts in years when the actual agronomic damage may be delayed by several weeks relative to the satellite-detected signal. Conversely, in years where the agronomic delay is real (confirmed by corrected SOS estimates), a correction-aware insurance index would reduce the false negative rate. Integrating the RiceBaCI-GEE correction layer into phenological index-based crop insurance frameworks for coastal Odisha and analogous Bay of Bengal delta regions has the potential to substantially reduce basis risk for the smallholder farmers who are most exposed to cyclone-associated yield losses.

Crop model data assimilation: Phenological dates derived from SAR-optical remote sensing are increasingly assimilated into process-based crop simulation models (DSSAT, ORYZA2000) to constrain simulated sowing dates, heading dates, and hence yield estimates (Mohite et al., 2019; Manikandan et al., 2025). Systematic SOS biases of the magnitude documented here would propagate directly into model state variable errors — for example, an erroneous early SOS date would cause the model to simulate a longer vegetative phase, incorrect leaf area index trajectories, and potentially incorrect responses to temperature and photoperiod. Mohite et al. (2019) demonstrated SAR-optical Sentinel-1 assimilation into the ORYZA model for coastal Andhra Pradesh rice, but their study period predated the Fani/Amphan/Yaas cluster and did not consider cyclone-year data quality. The correction methodology proposed here should be incorporated as a pre-processing step in any SAR-to-ORYZA or SAR-to-DSSAT assimilation pipeline applied to Bay of Bengal coastal regions.

5.4 Limitations

Several limitations of this study require transparent acknowledgement. First, the validation strategy is deliberately based on multiple independent open-data sources (MCD12Q2, ICRISAT VDSA, FAO–GIEWS calendars, district yield records) rather than dense in-situ BBCH-stage observations from a single agrometeorological network. This design maximises reproducibility and is methodologically defensible for cyclone-affected coastal regions where systematic field campaigns are unsafe and impractical, but it does not achieve the pixel-by-pixel spatial density of dedicated in-situ phenology observations. Future work integrating institutional ground-network data, where collaborative arrangements permit, would further refine the pixel-level error envelope; we treat that as a complementary line of work rather than a precondition for the present open-data framework. Second, the n = 480 classifier reference-label set used to train and evaluate the v0.3.0 saline-flood classifier is derived from automated multi-source provenance (§2.4: Copernicus EMS EMSR357 for Fani; UN-SPIDER (2019) Sentinel-1 pre/post change-detection for Amphan and Yaas; Sentinel-1∩WorldCover∩JRC mask for agronomic-flood negatives) rather than from analyst visual interpretation of high-resolution imagery. This choice replaces the originally pre-registered PlanetScope NICFI visual-interpretation pathway (OSF c4mp8 §E5; Planet Labs / KSAT TFO eligibility restricted to forest-domain users, 2026-05-06; academic appeal not answered within project SLA) and was selected over the §E5 Sentinel-2 visual-interpretation fallback because the automated provenance is fully reproducible from public products, independent of analyst judgement, and SHA-256-pinned in the Mendeley deposit. The principal limitation of this choice is that the cyclone-flood positive labels inherit the operational geolocation tolerance of EMSR357 (≈10 m horizontal) and the −3 dB Sentinel-1 change-detection threshold of UN-SPIDER (2019), both of which can miss thin surge-channel features that fall below the 10 m Sentinel-1/EMS resolution; agronomic-flood negative labels inherit the JRC seasonal-water mask's known under-detection of small bunded paddy patches. We treat both biases as conservative: they push the classifier OA downwards by design, so the reported OA = 0.844 SAR-only is, if anything, a lower bound on the true discrimination performance. Third, the GEE prototype implementation uses a 3×3 boxcar focal-mean for Sentinel-1 despeckling in Module 01, which is less effective at suppressing speckle whilst preserving edge features than the refined Lee sigma filter used in production. Whilst this limitation is noted in the code comments and the production modules apply the refined filter, any slight inconsistency between prototype and production pre-processing could affect the reproducibility of early exploratory results. Fourth, although the classifier achieves the pre-registered OA ≥ 0.88 / F1 ≥ 0.85 thresholds (achieved: OA = 0.990 full-feature, OA = 0.844 SAR-only on the v0.3.0 release) against the automated multi-source reference-label set (§2.4) on the Odisha study area and shows promising transferability to Andhra Pradesh, its applicability to other cyclone-exposed coastal deltas — the Irrawaddy Delta in Myanmar, the Mekong Delta in Vietnam, the Ganges–Brahmaputra–Meghna plain in Bangladesh — has not been tested. The classifier relies on IBTrACS North Indian Ocean basin cyclone tracks as a spatial feature, and its extension to other basins (Western Pacific, North Atlantic) would require the equivalent track data to be ingested as a GEE feature collection. Fifth, this study does not address the panicle initiation (PI) sub-stage, which is the most critical phenological checkpoint for determining cyclone-induced yield loss from saline stress after transplanting. Detection of PI from the red-edge spectral region (CIre, B5/B7 ratio) is treated as an exploratory analysis in this study and is not claimed as a confirmatory contribution.

5.5 Future Work

Several directions emerge naturally from the present findings. First, the classification of rice establishment method — distinguishing transplanted Puddled rice (TPR) from direct-seeded rice (DSR) — is a logical extension of the saline-flood classifier, as TPR and DSR produce distinct SAR backscatter dynamics around the transplanting/germination period (Fikriyah et al., 2019) that interact differently with the cyclone-surge signal. This distinction is particularly important for coastal Odisha, where the transition from TPR to DSR is accelerating under labour constraints and climate adaptation policies. Second, the detection of panicle initiation (PI) using the Sentinel-2 red-edge CIre index (Jha et al., 2025) would complete the phenological calendar from transplanting to maturity at the pixel level, enabling the calculation of cyclone-induced reductions in grain-filling duration — a key determinant of yield loss. Third, the integration of the corrected phenological products with DSSAT or ORYZA2000 models, as demonstrated by Mohite et al. (2019) and Manikandan et al. (2025) for non-cyclone contexts, would allow simulation of cyclone-induced yield loss distributions with uncertainty bounds directly propagated from the pixel-level bootstrap confidence intervals produced here. Fourth, systematic application of the RiceBaCI-GEE framework to all major Asian river deltas with documented cyclone exposure — the Irrawaddy, Mekong, Ganges–Brahmaputra–Meghna, Red River, and Chao Phraya — would produce the first multi-delta inventory of cyclone-induced SAR phenology bias, providing the evidence base for a community recommendation on best practice for remote sensing products in cyclone-exposed coastal rice systems.

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6. Conclusions

This study presents the first empirical characterisation and correction of the confound between cyclone-induced saline storm-surge inundation and agronomic transplanting flooding in Sentinel-1/2 Kharif rice phenology retrieval. Across five coastal Odisha districts, eight Kharif seasons (2017–2024), and three named cyclone events (Fani 2019, Amphan 2020, Yaas 2021), we demonstrate that the C-band SAR backscatter decrease produced by storm-surge inundation is observationally indistinguishable from the transplanting trough on which all published rice phenology algorithms depend. Failure to correct this confound introduces systematic SOS, POS, and EOS errors during cyclone years that substantially exceed normal interannual variability, generating misleading inferences about climate impacts on rice phenology in precisely the most cyclone-stressed districts.

The RiceBaCI-GEE framework resolves this through a multi-feature random-forest classifier fusing Sentinel-1 backscatter, Sentinel-2 spectral indices, JRC water permanence, and ERA5 meteorological data to separate the two inundation types at 10 m pixel level. Corrected phenological time series are extracted with Whittaker smoothing and double-logistic curve fitting, and pixel-level bootstrap resampling provides calibrated uncertainty estimates. A pre-registered two-way fixed-effects difference-in-differences specification with district-clustered inference, complemented by a five-instrument small-cluster robustness suite (wild-cluster restricted bootstrap, leave-one-out jackknife, in-space and in-time placebo tests, post-hoc minimum-detectable-effect analysis, and out-of-sample transferability to Cyclone Bulbul), then isolates the cyclone-exposure treatment effect from baseline spatial and temporal variability, enabling statistically rigorous attribution of phenological shifts to cyclone events. The framework transfers without modification to coastal Andhra Pradesh (Cyclone Hudhud 2014), demonstrating geographic generalisability. All GEE code, processed phenological rasters, and validation reference data are openly archived, facilitating direct adoption by the remote sensing community and integration into crop insurance index design and crop model assimilation pipelines serving the world's most cyclone-exposed rice farming regions.

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CRediT Author Contribution Statement

Contribution (CRediT taxonomy)	Supranab Panda	Sarat Chandra Sahu
Conceptualisation	Lead	Supporting
Methodology	Lead	Supporting
Software (GEE code, Python analysis)	Lead	—
Formal analysis	Lead	—
Investigation	Lead	Supporting
Resources	Supporting	Lead
Data curation	Lead	—
Writing — original draft	Lead	—
Writing — review & editing	Lead	Supporting
Visualisation	Lead	Supporting
Supervision	—	Lead
Project administration	Lead	Supporting
Funding acquisition	—	Lead

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the authors used Perplexity Computer (Anthropic Claude Sonnet) for literature review, study design assistance, manuscript drafting and code generation.

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The processed corrected and uncorrected Sentinel-1/2 Kharif rice phenological date rasters (SOS, POS, EOS, correction bias ΔD, bootstrap 95% confidence intervals) for all five coastal Odisha districts and eight Kharif seasons (2017–2024), together with the district BACI panel (n = 384), the saline-flood classifier label set (n = 480), the RF model card v0.3.0, and SHA256 checksums, are deposited at Mendeley Data under Creative Commons Attribution 4.0 licence (DOI: 10.17632/z3zxk4xy3c.1). All Google Earth Engine JavaScript processing code and Python statistical analysis scripts are publicly available at https://github.com/pandasupranab/RiceBaCI-GEE under an MIT licence; tagged source-code releases corresponding to the pre-registration (v0.1.0-prereg) and successive submission snapshots are permanently archived on Zenodo (concept DOI: 10.5281/zenodo.20024578; v0.1.1-prereg DOI: 10.5281/zenodo.20024579; v1.0.0-submission DOI: 10.5281/zenodo.20585636; v1.0.1-submission DOI: 10.5281/zenodo.20587316). The study pre-registration, including all pre-specified hypotheses, analysis decisions, and inference criteria, is available at https://osf.io/c4mp8 (DOI: 10.17605/OSF.IO/C4MP8). All n = 480 classifier reference labels (Copernicus EMS EMSR357 cyclone-flood points for Fani; UN-SPIDER (2019) Sentinel-1 pre/post change-detection points for Amphan and Yaas; Sentinel-1∩WorldCover∩JRC agronomic-flood points), together with coordinates, dates, cyclone-event provenance fields, and SHA-256 checksums, are openly redistributable and are included in full in the Mendeley Data record (labels_panel_n480.csv, labels_features_n480.csv) under CC-BY-4.0.

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End of manuscript draft (v1.0.1-submission, 2026-06-08). All empirical numbers in this draft are estimated from the real Sentinel-2 phenology panel (n = 64 district-year SOS/POS rows, n = 44 estimable EOS rows after right-censoring 20 post-cyclone cells, across 8 coastal/inland Odisha districts, 2017–2024) and the real v0.3.0 saline-flood random-forest classifier (n = 480 automated reference labels with provenance: 80 Copernicus EMS EMSR357 + 160 UN-SPIDER (2019) Sentinel-1 change-detection + 240 Sentinel-1∩WorldCover∩JRC mask; n_train = 384, n_test = 96; OA = 0.990 full-feature, OA = 0.844 SAR-only, 5-fold CV OA = 0.831). The v2.1 classifier-corrected pipeline produces a small but defensible attenuation of the district-aggregated DiD coefficient (({}): 15.289 → 15.108 d, () = −0.181 d; ({}): −3.587 → −3.677 d, () = −0.090 d; (_{}): 0 → −0.239 d after restoring 20 censored cells), consistent with the bounded pixel share of cyclone-surge inundation at the district-aggregation scale (per-district nonzero pixel-share across the five coastal treatment districts: Fani 0.10–2.52 %, Amphan 0.04–1.92 %, Yaas 0.42–7.21 %; Table S13a) and with the pre-registered prediction $\tau_{raw} > \tau_{corrected} > 0$ for SOS. All artefacts — code (GitHub v1.0.1-submission), software archive (Zenodo 10.5281/zenodo.20587316), data (Mendeley 10.17632/z3zxk4xy3c.1), pre-registration (OSF c4mp8) — are publicly accessible and version-pinned. Author names, affiliations, ORCID identifiers, OSF registration ID (c4mp8), GitHub repository URL, and Zenodo/Mendeley DOIs are filled in §6 and the title block.